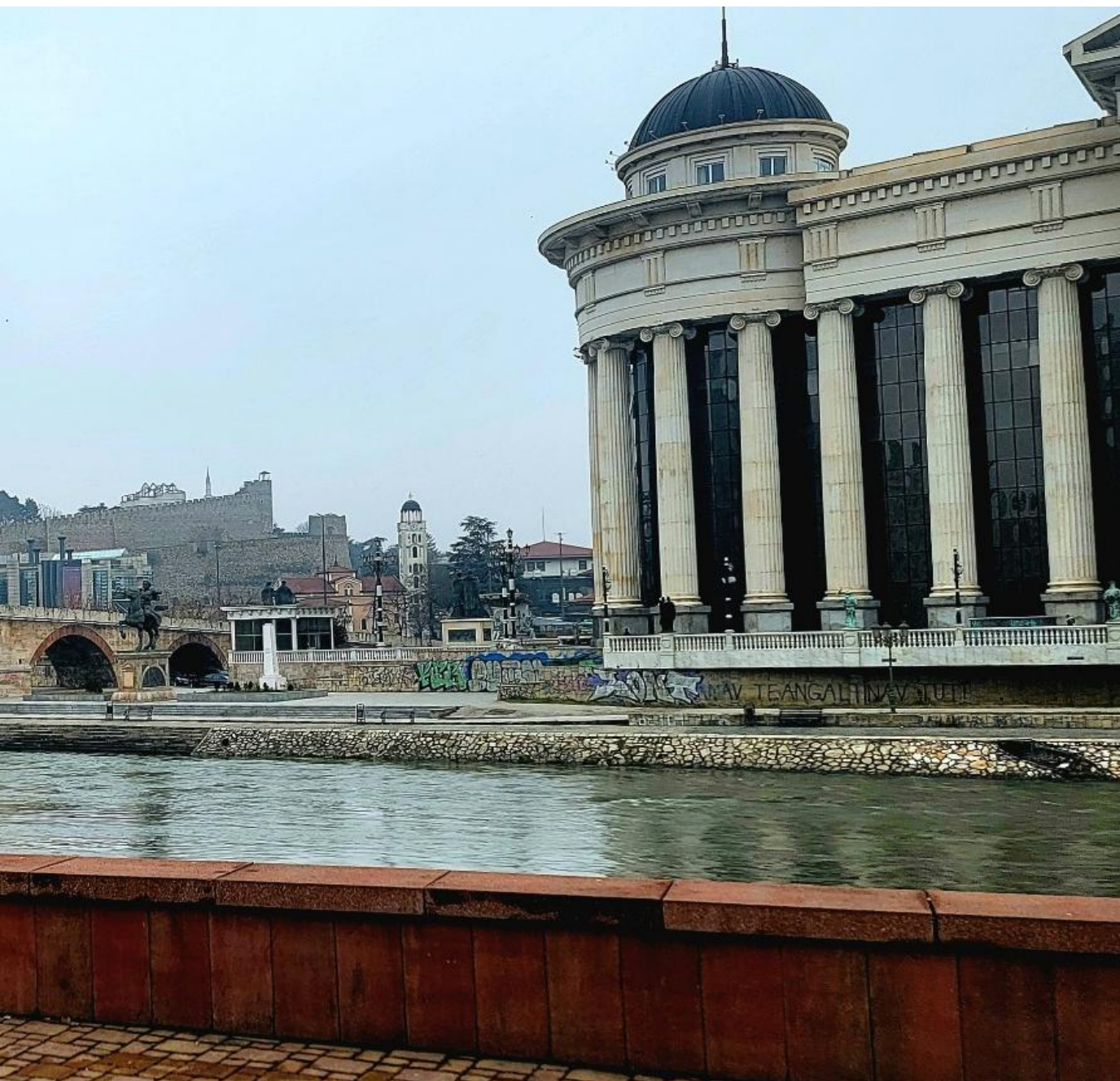


UNITED NATIONS ECONOMIC COMMISSION FOR EUROPE

**Validation of energy, climate, and air quality results  
under the Green Finance Facility to improve air quality  
and combat climate change in North Macedonia**



**UNECE**

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Geneva, 2026

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**Cut-off date for information used in this report is 26 December 2025.**

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## List of Abbreviations

|          |                                                                                                      |
|----------|------------------------------------------------------------------------------------------------------|
| AC       | Air Conditioning                                                                                     |
| COP      | Coefficient of Performance                                                                           |
| DHW      | Domestic Hot Water                                                                                   |
| EE       | Energy Efficiency                                                                                    |
| EF       | Emission Factor                                                                                      |
| EMEP/EEA | European Monitoring and Evaluation Programme / European Environment Agency                           |
| EU       | European Union                                                                                       |
| GFF      | Green Finance Facility to Improve Air Quality and Combat Climate Change in North Macedonia programme |
| GHG      | Greenhouse Gas                                                                                       |
| IPCC     | Intergovernmental Panel on Climate Change                                                            |
| IOM      | International Organization for Migration                                                             |
| NDC      | Nationally Determined Contribution                                                                   |
| PV       | Photovoltaic                                                                                         |
| QA       | Quality Assurance                                                                                    |
| RED      | Renewable Energy Directive (EU) 2018/2001                                                            |
| RES      | Renewable Energy Sources                                                                             |
| SDG      | Sustainable Development Goals                                                                        |
| SEC      | Specific Energy Consumption                                                                          |
| SI       | International System of Units                                                                        |
| SPF      | Seasonal Performance Factor                                                                          |
| UNECE    | United Nations Economic Commission for Europe                                                        |
| UNDP     | United Nations Development Programme                                                                 |



## Units Used

|                       |                                                                |
|-----------------------|----------------------------------------------------------------|
| CO <sub>2</sub>       | Carbon Dioxide                                                 |
| CO <sub>2</sub> e     | Carbon Dioxide Equivalent                                      |
| g                     | Gram                                                           |
| g/GJ                  | Grams per gigajoule                                            |
| kg                    | Kilogram                                                       |
| kWh                   | Kilowatt-hour                                                  |
| m <sup>2</sup>        | Square meter                                                   |
| m <sup>3</sup>        | Cubic meter                                                    |
| MW                    | Megawatt                                                       |
| MWh                   | Megawatt-hour (1 MWh = 1,000 kWh)                              |
| PM                    | Particulate Matter                                             |
| PM <sub>10</sub>      | Particulate Matter with aerodynamic diameter ≤ 10 micrometres  |
| PM <sub>2.5</sub>     | Particulate Matter with aerodynamic diameter ≤ 2.5 micrometres |
| tCO <sub>2</sub>      | Metric tons of carbon dioxide                                  |
| tCO <sub>2</sub> /MWh | Tons of CO <sub>2</sub> per megawatt-hour                      |
| tCO <sub>2</sub> e    | Metric tons of carbon dioxide equivalent                       |
| µg/m <sup>3</sup>     | Micrograms per cubic meter                                     |

## Abstract

This report presents a technical validation of energy, climate, and air-quality results reported under the Green Finance Facility (GFF) in North Macedonia, covering both Outcome 1 (SME and Industrial) and Outcome 2 (Households) portfolios. It validates the performance of 84 (of which 70 verified) industrial and 212 household investments, assessing estimated energy savings, greenhouse gas emission reductions, installed renewable energy capacity, and air quality benefits. The validation applies a dual-track methodology: a portfolio-level consistency check for industrial projects and a structured, project-level quality assurance process for households, both aligned with national rulebooks and international standards. Based on these validated results, the report consolidates programme-level impacts and provides an evidence-based update of targets for the 2024–2027 period, supporting transparent, consistent, and credible impact reporting for the remainder of the programme implementation.

# Executive Summary

This report presents a technical validation of energy, climate, and air-quality results reported under the Green Finance Facility to Improve Air Quality and Combat Climate Change in North Macedonia (GFF) programme funded by the UN Joint SDG Fund, covering Outcome 1 (SMEs) and Outcome 2 (households). The analysis verifies estimated energy savings, greenhouse gas (GHG) emission reductions, installed renewable energy (RES) capacity, and air quality (PM<sub>2.5</sub> and PM<sub>10</sub>) benefits for 2023-2025, and updates programme targets for the 2026–2027 period based on validated results and scenario-based sensitivity analysis.

All results presented in this report are based on verified project data extracted as of 26 December 2025, which serves as the cut-off date for the study.

The key findings and conclusions are summarized below:

## 1. Outcome 1 – portfolio-level performance confirmed

Validation at portfolio level confirms that reported energy savings, CO<sub>2</sub> reductions, and RES capacity from industrial investments are internally consistent and corresponding with the scale and structure of supported applications. The applied harmonized methodology is robust and suitable for continued programme-level reporting.

## 2. Outcome 1 – updated targets are evidence-based and realistic

Based on verified results from the initial implementation period, updated targets project annual impacts by 2027 of approximately 104,000 MWh of energy savings, 95,000 tons of CO<sub>2</sub> reductions, and around 70 MW of installed RES capacity, alongside measurable air-quality co-benefits.

## 3. Outcome 2 – Strong uptake of electrification technologies

The validation of 212 household investments confirms a strong dominance of high-efficiency heat pumps, reflecting a clear market shift toward electrification. This technology is the primary driver of energy savings, CO<sub>2</sub> reductions, and improvements in local air quality under Outcome 2.

## 4. Outcome 2 – Improved consistency in programme-level reporting

While project-level CO<sub>2</sub> reductions are based on verified, conservative calculations, the application of a weighted emission factor for programme-level projections improves internal consistency between energy and climate indicators without altering historical project results.

**5. Air quality co-benefits are significant across both outcomes**

Although not part of the original target framework, reductions in PM<sub>2.5</sub> and PM<sub>10</sub> emerge as important co-benefits, particularly under Outcome 2, supporting broader public health and environmental objectives.

**6. Sensitivity analysis confirms robustness of targets**

Scenario analysis indicates that programme performance remains robust under conservative assumptions. The principal risk relates to reduced investment volume, while technology mix shifts have a more limited impact at portfolio level.

**7. Clear basis for forward planning and reporting**

The validated results and updated targets provide a credible and transparent basis for results-based management, future reporting, and strategic decision-making for the remainder of the GFF programme period through 2027.

**Key recommendation:**

The GFF programme should continue applying the validated and harmonized methodologies for both outcomes, while formally embedding the updated targets, sensitivity ranges, and air-quality indicators into the core monitoring framework. This will ensure consistent, transparent, and decision-relevant reporting as the programme scales through 2027.

# Introduction

The Green Finance Facility to Improve Air Quality and Combat Climate Change in North Macedonia (GFF) is a UN Joint Programme funded by the UN Joint SDG Fund designed to improve air quality and combat climate change in North Macedonia by accelerating investments in energy efficiency and renewable energy. The GFF programme is implemented by the United Nations Development Programme (UNDP) Office in North Macedonia (lead), the International Organization for Migration (IOM), and the United Nations Economic Commission for Europe (UNECE) (implementing partners). Key partners in the GFF programme are the Government of North Macedonia, the European Bank for Reconstruction and Development (EBRD), and the local financial institutions in North Macedonia. The programme addresses structural challenges related to high household and enterprise energy consumption, widespread use of inefficient and polluting heating technologies, and limited access to affordable green finance — particularly for small and medium-sized enterprises (SMEs) and underserved households.

Through a combination of favorable financing, technical standards, and results-based monitoring, the GFF programme supports the deployment of clean heating systems, energy-efficient technologies, and renewable energy solutions. In doing so, the programme delivers measurable reductions in greenhouse gas emissions and particulate matter (PM<sub>2.5</sub> and PM<sub>10</sub>), while improving indoor comfort, lowering energy costs, and contributing to national and international climate and energy commitments.

The GFF programme is implemented through two complementary areas. Outcome 1 focuses on energy efficiency and renewable energy investments in SMEs and industrial enterprises, where project characteristics, production profiles, and baseline conditions vary significantly. Outcome 2 targets household-level investments, primarily related to space heating and domestic hot water systems, where interventions are more standardized and closely linked to residential energy use, affordability, and local air-quality impacts.

The study Validation Report of Energy, Climate, and Air Quality Results under the Green Finance Facility is funded by the UNECE Regular Programme of Technical Cooperation (RPTC). The preparation of this report was supported by the United Nations Development Programme (UNDP) Country Office in North Macedonia. The purpose of this report is twofold: first, to validate the results reported under both outcomes for the 2023–2025 implementation period; and second, to update programme targets for the 2026–2027 period based on validated performance trends. For Outcome 1, validation is conducted at the portfolio level, with emphasis on internal coherence, proportionality, and consistency across aggregated indicators. For Outcome 2, validation is also applied at the portfolio level, supported by a structured Quality Assurance process to confirm the

technical plausibility and methodological robustness of reported energy savings, emission reductions, and air-quality impacts prior to aggregation.

Beyond validation of past performance, the report applies a scenario-based sensitivity analysis to assess the robustness of updated targets under varying assumptions related to investment volume, technology mix, and realization rates. This approach supports the definition of realistic performance ranges, including conservative minimum thresholds and optimistic upper bounds, for the remainder of the programme period.

The findings of this report are intended to support UNDP and programme partners in strengthening results-based management of the GFF programme. By confirming the credibility of reported impacts, clarifying methodological boundaries, and identifying key risks and opportunities, the report provides a sound basis for transparent reporting, informed decision-making, and effective allocation of resources until 2027.

# 1. Methodology

This chapter presents the methodological framework used to validate the results and update the targets of the GFF programme across both programme outcomes. Given the fundamentally different nature of the supported interventions—ranging from portfolio-based SME and industrial investments under Outcome 1 to individual household solutions under Outcome 2—the validation approach is designed to ensure consistency, credibility, and proportionality while respecting the specific characteristics of each segment.

Accordingly, a Dual-Track validation methodology is applied. Outcome 1 is assessed at portfolio level, focusing on the coherence and plausibility of aggregated energy, climate, and air-quality impacts relative to investment volumes. Outcome 2 is validated at portfolio level through a structured quality-assurance process, ensuring technical realism and internal consistency of household-level results prior to aggregation. Together, these approaches provide a robust and transparent basis for confirming reported achievements and informing the revision of programme targets.

## 1.1 Overview<sup>1</sup>

This chapter defines the standardized methodological framework used to validate the results of the GFF programme and update the established targets. Given the distinct nature of the two programme segments, validation adopts a Dual-Track Approach to ensure internal consistency while respecting the differences between mass-market household investments and adapted industrial projects.

To ensure credibility and alignment with international climate reporting standards, the methodology is structured as follows:

- **Validation protocols (Dual-Track):**
  - *Outcome 1 (SME & Industrial):* Uses a Portfolio-Level Verification (Implied Factor Analysis) to cross-check aggregated impacts against investment volumes and regional baselines.
  - *Outcome 2 (Households):* Uses Portfolio-Level Quality Assurance (QA) to verify the technical plausibility of individual interventions against realistic performance bounds.

---

<sup>1</sup> NOTE: Heat pumps with COP greater than 4.5 are considered under renewable capacity.

- **GHG calculation framework:**
  - Aligned with IPCC Guidelines (2006/2019) and the *Rulebook on Energy Audits of the Republic of North Macedonia* (Official Gazette of the Republic of North Macedonia, No. 94/2013, 4 July 2013).
- **Air quality (PM) calculation framework:**
  - Aligned with the EMEP/EEA Air Pollutant Emission Inventory Guidebook (2019) for calculating particulate matter reductions from fuel switching.

## 1.2 Data Validation Protocol

All quantitative analyses presented in this report are based on a defined dataset extracted from the project's central Excel database as of 26 December 2025, which was established as the formal cut-off date for data processing and validation. The project database is subject to continuous updates as new applications are received, verified or corrected. Introducing a fixed cut-off date was therefore necessary to ensure consistency, traceability and reproducibility of the results presented. Any changes, additions or corrections made to the database after this date are not reflected in the figures, tables and maps included in this report and will be captured in subsequent reporting cycles.

For both Outcome 1 and Outcome 2, validation is conducted at the portfolio level, focusing on the consistency, proportionality, and plausibility of aggregated results rather than on individual sub-project performance.

### Outcome 1

Given the diversity of technologies and implementation approaches within the industrial portfolio, validation under Outcome 1 was carried out at portfolio level, focusing on the coherence, proportionality, and internal consistency of aggregated results rather than on the technical re-assessment of individual projects. This approach reflects the role of Outcome 1 in setting and tracking programme-level targets and achievements across heterogeneous energy efficiency (EE) and renewable energy (RES) investments.

#### Tier 1: Financial–energy consistency

At the first level, aggregated energy and capacity results were assessed in relation to total investment volumes using normalized intensity indicators. For energy efficiency projects, the primary indicator applied was EUR per annual MWh of energy savings, while renewable energy investments were assessed using EUR per MW of installed capacity and, where relevant, EUR per annual MWh generated. Projects combining EE and RES elements were reviewed using both



indicators and interpreted within intermediate ranges. This step ensured that reported portfolio-level impacts are commensurate with the scale of financial support provided.

- Energy Validation ( $IEF_{MWh}$ ):

$$IEF_{MWh} = \frac{\text{Verified Investment}}{\text{Verified MWh Savings}} \left[ \frac{\text{EUR}}{\text{MWh/yr}} \right]$$

$$IEF_{MW} = \frac{\text{Verified Investment}}{\text{Verified MW}} \left[ \frac{\text{EUR}}{\text{MW}} \right]$$

## Tier 2: Environmental impact intensity checks

To confirm consistency between energy outcomes and reported environmental benefits, portfolio-level impact intensities were calculated. These included EUR per ton of CO<sub>2</sub> reduced and EUR per gram of PM<sub>2.5</sub> and PM<sub>10</sub> reduced. The resulting ratios were reviewed across EE, RES, and combined portfolios to ensure that emissions and air-pollution reductions scale proportionally with energy impacts and reflect plausible average emission factors for the national industrial energy mix. This tier supports the identification of potential methodological inconsistencies without requiring technology solution-specific data.

- GHG Validation ( $IEF_{CO_2}$ ):

$$IEF_{CO_2} = \frac{\text{Verified Investment}}{\text{Verified tCO}_2} \left[ \frac{\text{EUR}}{\text{tCO}_2/\text{yr}} \right]$$

- PM Validation ( $IEF_{PM}$ ):

$$IEF_{PM} = \frac{\text{Verified Investment}}{\text{Verified g/yr Savings}} \left[ \frac{\text{EUR}}{\text{g/yr}} \right]$$

## Tier 3: Portfolio coherence and target alignment

Finally, aggregated results were reviewed for overall portfolio coherence, with particular attention to the relationship between planned targets and achieved results over time, as well as alignment between financial disbursement and delivered impacts. Portfolio segments with a disproportionate influence on total results were identified to inform interpretation of achievements and to support continuous improvement of future reporting cycles.

This tiered, portfolio-level validation approach provides a transparent and proportionate basis for confirming the credibility of Outcome 1 results, while remaining consistent with the level of detail available for industrial investments.

At portfolio level, the aggregated results translate into indicative investment and impact intensity ratios of approximately EUR 1,150 per MWh of annual energy savings, EUR 3,740 per ton of CO<sub>2</sub> reduced per year, and EUR 590 per kW of installed renewable capacity. These values are presented as descriptive indicators reflecting the scale and structure of the supported investments, rather than as performance benchmarks. Their magnitude is consistent with the diverse mix of industrial energy efficiency and renewable energy interventions supported under the programme and supports the overall plausibility of the reported targets and achievements.

## Outcome 2

The validation of 212 household projects was conducted through a structured Three-Tier Quality Assurance (QA) process, designed to ensure internal consistency, technical plausibility, and robustness of the reported results prior to aggregation at programme level.

### Tier 1: Unit consistency and outlier screening

At the first level, reported energy savings were screened for unit consistency and scale plausibility to ensure comparability across household assessments. During this step, values outside the expected range for typical residential installations were identified for further review, as unusually high figures may indicate differences in unit scaling or reporting methodologies.

- **Trigger condition:** Reported annual energy savings exceeding 25 MWh/year per household.
- **QA action:** Values exceeding this threshold were flagged for review and, where appropriate, standardized to ensure consistency of magnitude across the dataset.
- **Verification:** Reviewed values were cross-checked against the approved loan amount and the installed technology type to confirm alignment between the scale of investment and the reported performance.

This step ensured that all household results were expressed on a harmonized and comparable basis prior to aggregation.

### Tier 2: Technical plausibility check

Following unit consistency screening, reported savings were assessed against the physical and operational limits of the installed technologies to confirm technical plausibility and prevent over-estimation at portfolio level.

- **Heat pumps:** Reported savings were assessed based on realistic seasonal performance ranges, assuming a maximum practical Coefficient of Performance (COP) of more than 4.5, in line with climatic conditions in North Macedonia. Any implied Seasonal Performance Factor (SPF) exceeding 5.0 was flagged for manual review.

- **Biomass boilers:** Efficiency assumptions were assessed using conservative real-world benchmarks, with upper efficiency limits aligned at 90 per cent (lower heating value), reflecting typical operating conditions rather than nominal manufacturer values.

This tier ensured that reported impacts remained within credible engineering boundaries.

### **Tier 3: Sampling strategy for independent verification**

To complement the desk-based validation, an on-site verification plan has been defined as covering 10 per cent of beneficiary households. The sample is stratified to reflect the relative impact and risk profile of different technologies within the portfolio:

- **High-impact technologies (heat pumps):** 65 per cent of the sample, reflecting their dominant contribution to energy and emissions outcomes.
- **Higher pollution-risk technologies (biomass systems):** 20 per cent of the sample.
- **Long-term passive solutions (insulation and window replacement):** 15 per cent of the sample.

This tier provides an additional layer of assurance on installation quality and operational performance, supporting the credibility of programme-level reporting.

## **1.3 Greenhouse Gas (GHG) Emission Calculation Methodology**

### **Standard and calculation**

To ensure internal consistency for target-setting within this report, a standardized Weighted Emission Factor (0.71 tCO<sub>2</sub>/MWh) was applied only for the purpose of indicative projections. A dedicated programme-level CO<sub>2</sub> accounting methodology is additionally developed to guide future consultant reporting and ensure a consistent, transparent approach across technologies and projects.

GHG reductions are calculated using an Activity-Based Approach consistent with IPCC Tier 2 methods. The methodology focuses on the avoided emissions principle, comparing the emissions of the baseline heating system against the new efficient technology.

Formula:  $GHG_{avoided} = \sum(E_{baseline,i} \times EF_{fuel,i}) - \sum(E_{project,i} \times EF_{fuel,i})$

Where:

- $GHG_{avoided}$ : Total annual emissions reduced (tCO<sub>2</sub> eq/year).
- $E_{baseline,i}$ : Annual energy consumption for the specific fuel/technology (MWh/year).

- $EF_{fuel,i}$ : Emission Factor for the specific fuel source (tCO<sub>2</sub> eq/MWh).

## Emission factors

To ensure local relevance and compliance with national legislation, emission factors are sourced directly from the Rulebook on Energy Audits of the Republic of North Macedonia (Official Gazette of the Republic of North Macedonia, No. 94/2013, 4 July 2013).

Table 1 presents the emission factors utilized in the modeling of the CO<sub>2</sub> emissions for the relevant energy sources.

Table 1. CO<sub>2</sub> emission factors

| Energy Source    | Emission (t/MWh) | Factor | Justification                                                                                                                                                                                                                |
|------------------|------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grid Electricity | 0.915            |        | Reflects the high carbon intensity of the North Macedonian grid (dominated by lignite generation at REK Bitola).                                                                                                             |
| Biomass (Wood)   | 0/0.403*         |        | While often treated as "carbon neutral" in EU ETS, for household heating in North Macedonia involving non-sustainable harvesting and inefficient combustion, Tier 2 factor is applied to avoid overstating climate benefits. |
| Fossil Fuels     | 0.267            |        | Standard factor for heating oil/liquid fuels.                                                                                                                                                                                |

**NOTE:** The CO<sub>2</sub> emission is 0 kg CO<sub>2</sub>/kWh, a value published in the Rulebook on Energy Audits; however, according to the "IPCC Guidelines, 2006, National Greenhouse Gas Inventory, Volume 2: Energy," a value of 0.403 kg CO<sub>2</sub>/kWh for wood/wood waste is indicated, which has been taken into account in the calculations.

## Weighted Portfolio Factor

For programme-level projections and target setting, a *Weighted Emission Factor* of 0.71 tCO<sub>2</sub>/MWh was derived. This factor is a weighted average reflecting the actual baseline heating mix of the 212 validated households:

- 64 per cent Baseline: Electric Heating (0.915 tCO<sub>2</sub> /MWh).
- 19 per cent Baseline: Biomass/Wood (0.403 tCO<sub>2</sub> /MWh).
- 17 per cent Baseline: Fossil Fuels (0.267 tCO<sub>2</sub> /MWh).

$$EF_{weighted} = (0.64 \times 0.915) + (0.19 \times 0.402) + (0.17 \times 0.267) \approx 0.71 \text{ tCO}_2 / \text{MWh}$$

This weighted factor provides a conservative and statistically representative metric for projecting future impact.

## 1.4 Particulate Matter (PM) Calculation Methodology

### Standard and calculation

Reductions in PM<sub>2.5</sub> and PM<sub>10</sub> are calculated according to the EMEP/EEA Air Pollutant Emission Inventory Guidebook 2019 (Chapter 1.A.4 Small Combustion). This methodology quantifies the reduction in local air pollution resulting from fuel switching (e.g., from raw wood to electricity or efficient pellets).

$$\text{Formula: } PM_{reduced} = (E_{thermal} \times EF_{PM,baseline}) - (E_{thermal} \times EF_{PM,new})$$

### PM Emission factors

The calculation accounts for the dramatic difference in local emissions between solid fuel combustion and electrified heating.

Table 2 presents the emission factors utilized in the modeling of the PM emissions for the relevant energy sources.

Table 2. PM<sub>2.5</sub> and PM<sub>10</sub> factors

| NFR Code  | Technology                                  | Fuel                        | Pollutant         | Emission Factor | Unit | Range   | Guidebook Table |
|-----------|---------------------------------------------|-----------------------------|-------------------|-----------------|------|---------|-----------------|
| 1.A.4.b.i | Conventional boilers < 50 kW                | Wood and similar wood waste | PM <sub>2.5</sub> | 470             | g/GJ | 235–940 | Table 3-43      |
| 1.A.4.b.i | Conventional boilers < 50 kW                | Wood and similar wood waste | PM <sub>10</sub>  | 480             | g/GJ | 240–960 | Table 3-43      |
| 1.A.4.b.i | Pellet stoves and boilers (burning pellets) | Wood pellets (biomass)      | PM <sub>2.5</sub> | 30              | g/GJ | –       | Table 3-49-06   |

|                  |                                             |                        |                   |     |      |         |               |
|------------------|---------------------------------------------|------------------------|-------------------|-----|------|---------|---------------|
| <b>1.A.4.b.i</b> | Pellet stoves and boilers (burning pellets) | Wood pellets (biomass) | PM <sub>10</sub>  | 30  | g/GJ | –       | Table 3-49-06 |
| <b>1.A.4.b.i</b> | NA                                          | 'Other' Liquid Fuels   | PM <sub>2.5</sub> | 1.9 | g/GJ | 1.1-2.6 | Table_3-5     |
| <b>1.A.4.b.i</b> | NA                                          | 'Other' Liquid Fuels   | PM <sub>10</sub>  | 1.9 | g/GJ | 1.1-2.6 | Table_3-5     |

**Note:** PM reductions are claimed only for households switching from solid fuels (wood/coal). Households switching from old electric heaters to heat pumps are assigned zero local PM reductions, as the baseline has no on-site exhaust. Grid-level PM emissions are filtered at the power plant and not counted in local exposure.

## 1.5 Uncertainty Management Strategy

To account for inherent uncertainties in long-term forecasting (through 2027), a scenario-based modelling approach was integrated into the target-setting process. This approach stress-tests the "Base Case" projections against deviations in key input variables, specifically:

- **Market volume:** Variations in application rates.
- **Technology mix:** Shifts in user preference between heat pumps and biomass.
- **Operational performance:** Behavioral "take-back" effects reducing realized savings.

*Detailed definitions of these scenarios and their quantitative impacts are presented in Chapter 4: Sensitivity Analysis.*

## 1.6 Limitations

While this validation applies to rigorous statistical and engineering checks, the following limitations apply to the findings:

- **Lack of pre-retrofit metering:** As baseline energy consumption was not physically metered prior to intervention, savings are calculated based on "Expected Savings" methodologies (engineering estimates) rather than "Metered Savings" (ex-post measurement).

- **Behavioral variance:** The analysis assumes standard occupancy patterns. Actual savings may vary due to individual user behavior (e.g., window opening habits, thermostat preferences), which is accounted for via the "Rebound Effect" sensitivity scenario.
- **Grid factor stability:** CO<sub>2</sub> calculations assume a constant grid emission factor (0.915 tCO<sub>2</sub>/MWh). Significant decarbonization of the national electricity mix before 2027 would lower the calculated avoided emissions from heat pumps.

## 2. Overview of Implemented Projects and Technology Uptake

### Outcome 1

#### 2.1 Technology Mix Across All Supported Companies

The SMEs portfolio supported under Outcome 1 comprises a diverse combination of energy efficiency (EE) and renewable energy (RES) investments, reflecting the varying operational needs, baseline conditions, and investment capacities of participating SMEs. Based on the classification of supported projects, 64 per cent of investments are primarily renewable energy–focused, 23 per cent are dedicated energy efficiency interventions, and 13 per cent combine EE and RES technology solutions within a single project, as shown in Figure 1.

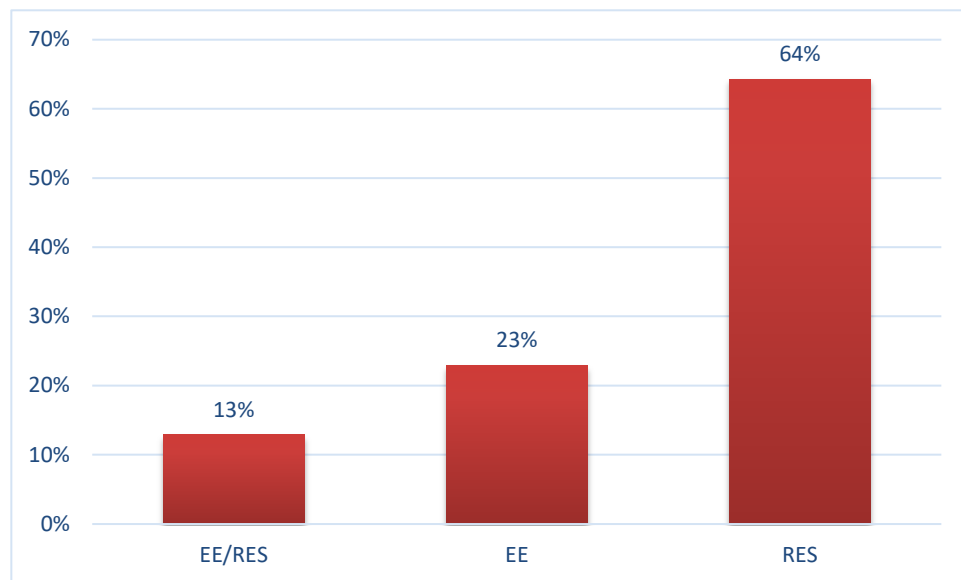


Figure 1. Distribution of projects by primary technology category - Outcome 1

#### Interpretation of the technology mix

##### Renewable energy as the dominant investment area

Renewable energy investments account for the largest share of the portfolio. This reflects strong interest from industrial beneficiaries in on-site energy generation as a means to reduce exposure to energy price volatility, improve energy security, and lower operating costs. The importance of RES projects also suggests that such investments are relatively well understood by companies and are often easier to justify internally due to their visible and measurable outputs, such as installed capacity and on-site electricity production.



### **Energy Efficiency: A Strategic but secondary component**

Projects implementing only energy efficiency technologies represent just under one quarter of the portfolio. While EE solutions typically offer higher long-term returns and system-wide benefits, their implementation in industrial settings often requires deeper process integration, detailed audits, and operational adjustments. These factors can act as barriers, particularly for smaller enterprises or for companies seeking quicker, more tangible investment outcomes. Nevertheless, EE interventions remain critical for improving overall system performance and for ensuring that renewable energy installations are appropriately sized and effectively utilized.

### **Combined EE and RES projects: Integrated approaches**

Projects combining energy efficiency and renewable energy elements account for 13 per cent of the portfolio. Although smaller in number, these integrated projects often deliver higher overall system performance by simultaneously reducing energy demand and supplying clean energy.

When considering total technology solutions implemented and excluding mixed EE–RES solutions as a separate category, energy efficiency–related interventions account for approximately 33 per cent of the portfolio, while renewable energy–related investments account for approximately 77 per cent.

### **Implications for programme design**

The observed technology mix suggests that future programme phases could benefit from:

- stronger integration of energy efficiency components within renewable energy projects,
- targeted advisory support to help enterprises identify and implement process-level efficiency solutions, and
- continued promotion of combined EE–RES solutions that maximize energy, climate, and economic benefits.

Overall, the technology mix under Outcome 1 reflects a pragmatic investment pattern in the industrial sector, with a clear preference for renewable energy deployment complemented by targeted efficiency improvements where conditions allow, as presented in Figure 2.

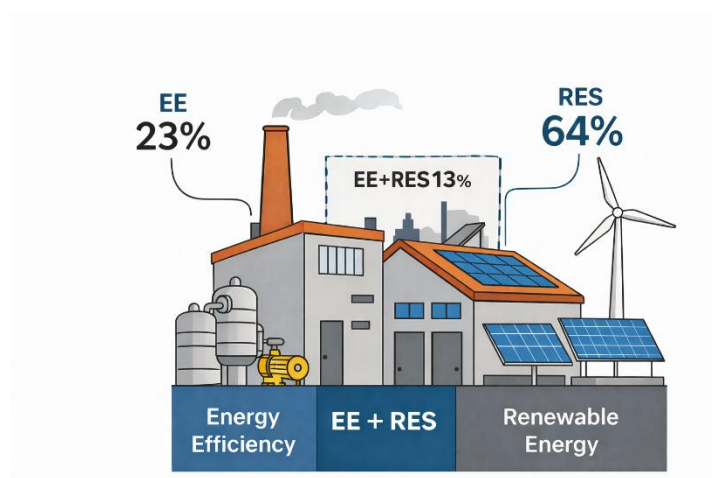


Figure 2. Technology distribution across the Outcome 1 portfolio<sup>2</sup>

## 2.2 Initial Targets and Actual Achievements – Outcome 1 (Validated Baseline)

Following the application of the Tier 1 Data Validation Protocol, unit consistency checks were applied to harmonize reported values and establish a robust and comparable baseline. Table 3 presents the validated performance of the 84 companies, of which 70 companies with verified results.

### Original Outcome 2 targets

- **Energy savings:** 81,000 MWh/year
- **CO<sub>2</sub> emission reductions:** 75,654 t/year
- **RES capacity:** 8.1 MW

Table 3. Actual state - Targets and achievements – Outcome 1

| Indicator                              | Achieved Value (expected) | Achieved Value (verified) | Target | Achievement (%) |
|----------------------------------------|---------------------------|---------------------------|--------|-----------------|
| Energy savings (MWh)                   | 65,345                    | 56,656                    | 81,000 | 70%             |
| CO <sub>2</sub> reductions (t/yr)      | 50,701                    | 39,610                    | 75,654 | 52%             |
| RES capacity (MW)                      | 42.30                     | 33.66                     | 8.10   | 416%            |
| PM <sub>2.5</sub> reductions (g/MWh/y) | 190                       | -                         | -      | -               |

<sup>2</sup> Source: Author's own illustration.

|                                       |     |   |   |   |
|---------------------------------------|-----|---|---|---|
| PM <sub>10</sub> reductions (g/MWh/y) | 354 | - | - | - |
|---------------------------------------|-----|---|---|---|

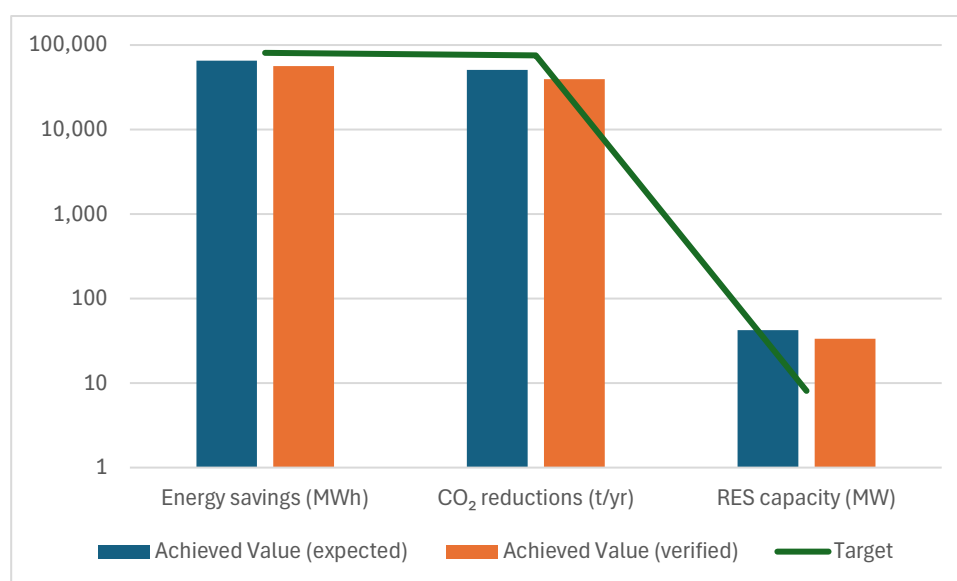


Figure 3. Initial targets vs. achievements – Outcome 1

Figure 3 illustrates the relationship between initial targets and achieved values for Outcome 1.

These results represent the actual performance of the programme to date and form the basis for the validation analysis in the following chapter. They reflect the true impact of the implemented technology solutions once unit consistency checks were applied to harmonize reported values and establish a robust and comparable baseline.

### 2.3 Updated Targets for Outcome 1 (2024–2027)

The validation of the first 84 SMEs, of which 70 SMEs with verified results, provides a solid empirical foundation for establishing realistic and meaningful targets for the rest of the programme period. When the original Outcome 1 targets were set, there was limited information on how SMEs would respond to the loan facility, which technologies they would prefer, or how large the resulting impacts would be. Now that two full implementation years are complete, the observed results allow for a recalibration of expectations based on evidence rather than assumptions.

The revised targets presented in this chapter are built on two core components:

1. Projected number of financed SMEs through 2027
2. Expected/validated average impact per SME, derived from 84 SMEs, of which 70 SME projects with verified results.

The programme began with one financed project in 2023 and expanded rapidly to 45 in 2024 and 41 in 2025. A moderate growth scenario for the coming years – 46 companies in 2026 and 17 in 2027 – results in a total of 150 supported projects over the 2023–2027 period. Applying the validated average impacts to this trajectory produces a new, evidence-based set of annual and cumulative targets, as presented in Table 4.

Table 4. Original Outcome 1 Targets vs. Updated Targets (2024–2027)

| Indicator                                   | Original Target | Updated Target (2024–2027) | Comment                                                                                                                                 |
|---------------------------------------------|-----------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Energy savings (MWh/year)</b>            | 81,000          | <b>104,000</b>             | Increased target reflecting the scale and composition of the industrial portfolio.                                                      |
| <b>CO<sub>2</sub> reductions (t/year)</b>   | 75,654          | <b>95,000</b>              | Based on verified project-level reductions; the implied average emission factor is consistent with the portfolio-level weighted factor. |
| <b>RES capacity (MW)</b>                    | 8.1             | <b>70</b>                  | Higher target driven by the strong share of RES and EE+RES projects.                                                                    |
| <b>PM<sub>2.5</sub> reduction (g/MWh/y)</b> | –               | <b>1,103 kg/y</b>          | Newly introduced indicator reflecting improved air-quality quantification.                                                              |
| <b>PM<sub>10</sub> reduction (g/MWh/y)</b>  | –               | <b>2,054 kg/y</b>          | Newly introduced indicator reflecting improved air-quality quantification.                                                              |

### Basis for updated calculations

#### Average per-household impacts

Using the validated dataset from 70 SMEs, the following average annual impacts were confirmed:

- **Energy savings:** 810.34 MWh/project,
- **CO<sub>2</sub> emission reduction:** used verified reduction (average factor 0.7 t/MWh), identical to the introduced weighted factor,
- **RES capacity:** based on average intensity observed in the portfolio (0.48 MW/project), and
- **PM reductions:**

- PM<sub>2.5</sub>: 2.81 g/MWh/y,
- PM<sub>10</sub>: 5.22 g/MWh/y

These averages form the basis for projecting future outcomes, providing a representative benchmark for estimating expected impacts, as presented in Table 5.

*Table 5. Updated Annual Targets – Outcome 1*

| Year                        | Applica<br>tions | Energy<br>Savings<br>(MWh/year) | CO <sub>2</sub><br>Reductions<br>(t/year) | RES<br>Capacity<br>(MW/year) | PM <sub>2.5</sub><br>Reductions<br>(g/MWh/y) | PM <sub>10</sub><br>Reductions<br>(g/MWh/y) |
|-----------------------------|------------------|---------------------------------|-------------------------------------------|------------------------------|----------------------------------------------|---------------------------------------------|
| 2023                        | 1                | 2,000                           | 1,500                                     | 0.2                          | 7                                            | 14                                          |
| 2024                        | 45               | 33,000                          | 30,000                                    | 20                           | 331                                          | 616                                         |
| 2025                        | 41               | 24,000                          | 15,000                                    | 18                           | 301                                          | 561                                         |
| 2026                        | 46               | 34,000                          | 30,000                                    | 22                           | 338                                          | 630                                         |
| 2027                        | 17               | 11,000                          | 18,500                                    | 10                           | 125                                          | 233                                         |
| <b>Total 2023–<br/>2027</b> | <b>150</b>       | <b>104,000</b>                  | <b>95,000</b>                             | <b>70</b>                    | <b>1,103</b>                                 | <b>2,054</b>                                |

Taking together, these updated targets present a much clearer picture of the programme's expected contribution by 2027 at portfolio level. Energy savings reach approximately 104,000 MWh per year, exceeding the original target and reflecting the scale and composition of the supported industrial investments. CO<sub>2</sub> reductions follow this trend, reaching around 95,000 tons per year, based on verified project-level results, with the implied average emission factor fully consistent with the portfolio-level weighted factor.

Renewable energy capacity increases substantially, reaching a cumulative 70 MW, driven by the strong predominance of RES-oriented and combined EE–RES projects. Although not originally targeted, reductions in PM<sub>2.5</sub> and PM<sub>10</sub> emerge as relevant co-benefits, with expected annual reductions of approximately 1,103 kg of PM<sub>2.5</sub> and 2,054 kg of PM<sub>10</sub> by 2027.

These updated targets offer a realistic and evidence-based direction for the remainder of the programme period and form a strong basis for reporting, planning, and resource allocation.

## 2.4 Strategic Implications

The validation of portfolio-level results and the definition of updated targets for Outcome 1 point to three key strategic implications for the remainder of the programme:

### **1. Consolidating renewable energy deployment:**

The dominance of RES-oriented projects within the industrial portfolio reflects a strong preference for on-site energy generation as a response to energy price instability and supply security concerns. Continued support for renewable energy investments is essential, as they are the primary contributors to the programme's RES capacity target and a significant driver of overall emissions reductions.

### **2. Strengthening the role of energy efficiency:**

While energy efficiency projects form a smaller share of stand-alone investments, their contribution becomes more significant when combined EE–RES projects are taken into account. This highlights the importance of promoting integrated approaches that reduce demand alongside increasing renewable supply. Targeted technical assistance and advisory support could help unlock additional efficiency potential, particularly in process-driven industrial environments.

### **3. Ensuring consistency between energy and climate reporting:**

Portfolio-level analysis confirms alignment between verified CO<sub>2</sub> reductions and energy savings, with the implied average emission factor consistent with the weighted factor applied for interpretation. Maintaining this consistency is critical for credible reporting, particularly as the technology mix evolves. Ongoing monitoring of the balance between EE and RES investments will help ensure that energy, climate, and air-quality objectives remain mutually reinforced.

## **Outcome 2**

The Green Finance Facility to Improve Air Quality and Combat Climate Change in North Macedonia Programme has successfully supported 212 individual investments, demonstrating strong market demand. The portfolio is heavily aligned toward high-efficiency electrification, with Heat Pumps accounting for 47 per cent of all installations. The scale of participation demonstrates a strong willingness among households to improve comfort, reduce energy costs, and transition toward cleaner heating and hot-water solutions when accessible financing is available.

### **2.5 Technology Mix Across All Supported Households**

A set of 212 financed applications cover a wide range of technologies. Their distribution is shown in Table 6.

Table 6. Technology Distribution – Outcome 2

| Technology solution              | Number of technology solutions applied |
|----------------------------------|----------------------------------------|
| Heat pumps                       | 136                                    |
| Biomass boilers                  | 45                                     |
| Window replacement               | 30                                     |
| Insulation                       | 23                                     |
| Domestic Hot Water (DHW) systems | 21                                     |
| Air-conditioning units           | 15                                     |
| Solar DHW systems                | 9                                      |
| Photovoltaic (PV) systems        | 9                                      |

- **Heat pumps (136):** The dominant technology, reflecting high user awareness and competitive running costs.
- **Biomass boilers (45):** Represents transitional technology for areas with weak grid reliability or strong cultural preference for fire-based heating.
- **Envelope solutions (53):** Insulation and window replacements remain limited in uptake (23 per cent), indicating a need for bundled "Envelope + Heat Pump (+PV)" incentives.
- **Solar thermal and PV systems** remain comparatively uncommon

Table 7 and Table 8 provide an overview of the technological solutions supported under Outcome 2. Table 7 presents the total number of implemented technological solutions, reflecting cases where a single application includes more than one solution, while Table 8 shows the distribution of applications by primary technology category. The comparison highlights the strong uptake of both renewable energy (RES) and energy efficiency (EE) solutions, as well as a significant share of combined interventions.

Table 7. Types of technological solutions

| Types of technological solutions | Number of solutions implemented |
|----------------------------------|---------------------------------|
| RES                              | 154                             |
| EE                               | 134                             |

Table 8. Distribution of applications by primary technology category

| Distribution of applications by primary technology solution category | Number of applications |
|----------------------------------------------------------------------|------------------------|
| RES                                                                  | 101                    |
| EE                                                                   | 66                     |
| Combination of both                                                  | 45                     |

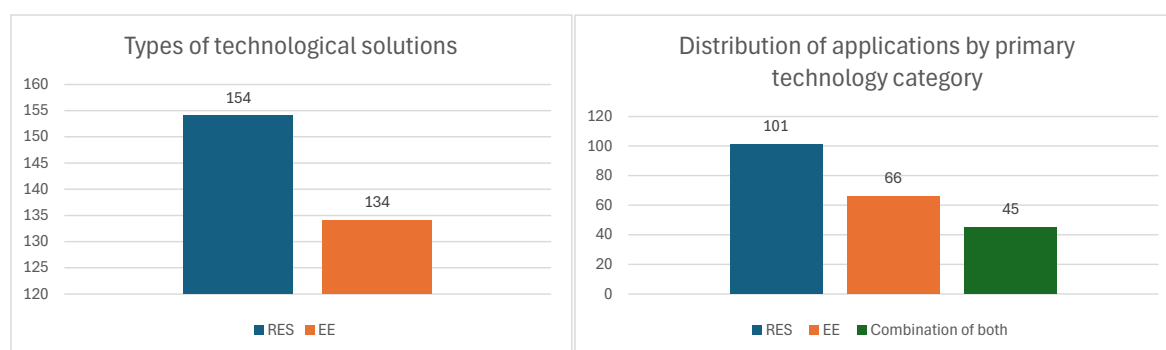


Figure 4. Distribution of technological solutions by type

Figure 4 provides complementary insights into the technological profile of the supported household investments. Figure 4 (left) shows the distribution based on the **total number of technological solutions installed**, where renewable energy (RE) technologies clearly dominate. This reflects the strong uptake of renewable-based systems, particularly heat pumps, which are often implemented together with other supporting solutions. As a result, a single household investment may include more than one technological solution and is therefore counted multiple times in this figure.

Figure 4 (right) presents the distribution **by number of applications**, where each application is counted once and classified according to its **primary technology category**. From this perspective, RES remains the most frequently selected main intervention. However, a notable share of applications falls under the combined energy efficiency (EE) and RES category, indicating that many households chose an integrated approach. This suggests that beneficiaries are increasingly combining building envelope improvements or efficiency solutions with renewable energy systems in order to achieve higher energy savings, greater emission reductions, and improved overall performance of their investments.

Overall, Figure 4 highlights a shift away from isolated solutions toward more comprehensive, bundled solutions, which align well with the programme's objectives of maximizing energy, climate, and air quality impacts at the household level.



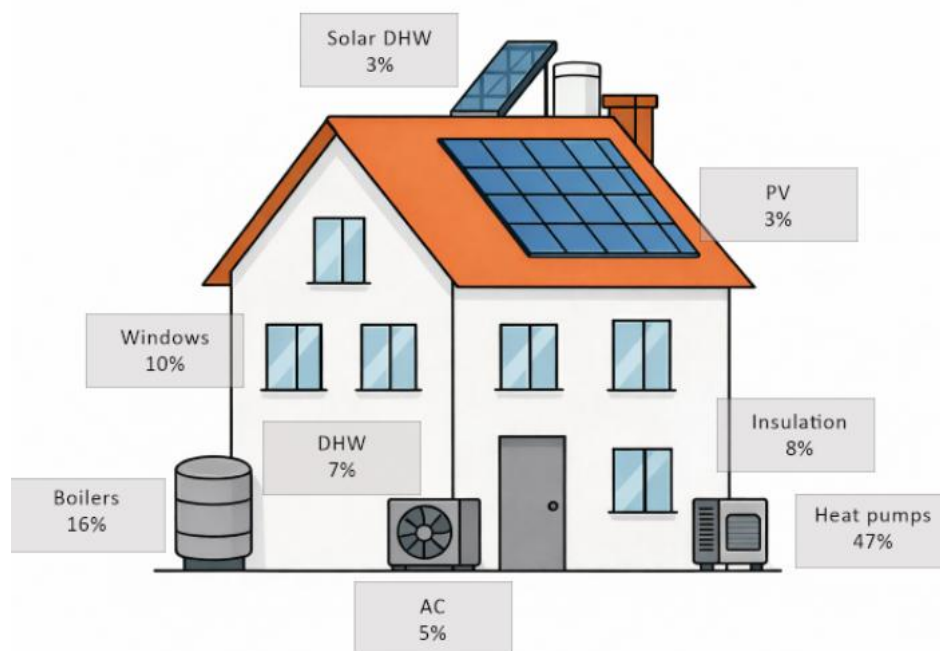


Figure 5. Technology distribution across the Outcome 2 portfolio<sup>3</sup>

Figure 5 illustrates the distribution of technological solutions across the Outcome 2 portfolio, with heat pumps representing the dominant technology, complemented by insulation, boilers, windows, and renewable energy systems.

### Interpretation of technology trends

#### Heat Pumps as the leading technology

With 136 installations, heat pumps have become the most widely adopted household solutions in the programme. Their popularity is consistent with broader trends in the region, where heat pumps are increasingly recognized for their:

- high energy efficiency,
- ability to reduce monthly heating expenses,
- contribution to cleaner indoor and outdoor air, and
- straightforward installation process.

The dominance of this technology suggests that awareness of its benefits is already relatively high among households.

<sup>3</sup> Source: Author's own illustration.

### **Biomass boilers: A Transitional Step**

The programme also includes 45 biomass boilers, indicating that a portion of households continue to rely on biomass as a preferred, familiar heating option or support system for heat pumps. For homes that previously used inefficient wood stoves or older boilers, switching to modern biomass systems provides substantial improvements in comfort, safety, and local air quality. These systems operate with much higher combustion efficiency and significantly lower particulate emissions compared to traditional stoves.

In several households, biomass boilers are installed together with heat pumps as part of a hybrid heating strategy. In these cases, the heat pump covers the majority of the heating season, while the biomass system is used only during the coldest days when temperatures fall below the optimal operating range of heat pumps. This approach reduces peak electricity demand, improves overall system resilience, and allows households to install heat pumps of smaller capacity, which often increases seasonal efficiency and lowers investment costs.

**However, the continued presence of biomass – even in modern form – also highlights the need for ongoing awareness raising on fully clean heating technologies and the long-term environmental benefits of transitioning to systems that eliminate combustion entirely, wherever feasible and economically viable.**

### **Envelope solutions: A notable gap**

Only 23 insulation solutions and 30 window replacements were financed. Despite offering some of the most durable long-term energy savings, these solutions represent a smaller share of investments. This may signal:

- technology solution already implemented (due to the necessity of minimum requirements for new and reconstructed buildings)
- limited awareness of the savings potential
- higher upfront costs compared with heating equipment
- preference for solutions that provide immediate comfort benefits.

Given their significant contribution to reducing heating demand, envelope solutions present a clear opportunity for targeted communication and advisory support.

### **Solar technologies: Still limited in uptake**

Solar DHW systems (nine installations) and PV systems (nine installations) remain underrepresented. These technologies offer strong long-term benefits, including reduced energy

bills and renewable energy generation, but they appear to require additional promotion or more favourable financing conditions to reach broader uptake.

Figure 6. Share of each technology in the total portfolio – Outcome 2

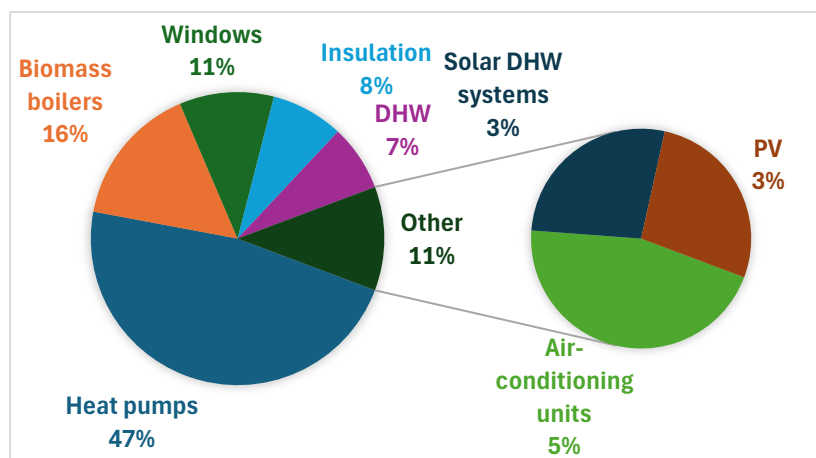


Figure 6 illustrates the relative contribution of each technology to the total Outcome 2 portfolio, with heat pumps accounting for the largest share, followed by biomass boilers, windows, and insulation measures.

## 2.6 Initial Targets and Actual Achievements – Outcome 2 (Validated Baseline)

Following the application of the Tier 1 Data Validation Protocol, unit consistency checks were applied to harmonize reported values and establish a robust and comparable baseline. Table 9 presents the validated performance of the 212 households.

### Original Outcome 2 targets

- **Energy savings:** 5,200 MWh/year
- **CO<sub>2</sub> emission reductions:** 4,857 t/year
- **RES capacity:** 2.6 MW

Table 9. Actual state – targets and achievements – Outcome 2

| Indicator                         | Achieved Value | Target | Achievement (%) |
|-----------------------------------|----------------|--------|-----------------|
| Energy savings (MWh/yr)           | 3,308          | 5,200  | 64%             |
| CO <sub>2</sub> reductions (t/yr) | 2,349          | 4,857  | 48%             |
| RES capacity (MW)                 | 1.94           | 2.6    | 75%             |

|                                            |       |   |   |
|--------------------------------------------|-------|---|---|
| <b>PM<sub>2.5</sub> reductions (kg/yr)</b> | 2,550 | – | – |
| <b>PM<sub>10</sub> reductions (kg/yr)</b>  | 4,498 | – | – |

These results represent the actual performance of the programme to date and form the basis for the validation analysis in the following chapter. They reflect the true impact of the implemented solutions once unit consistency checks were applied to harmonize reported values and establish a robust and comparable baseline.

## 2.7 Updated Targets for Outcome 2 (2024–2027)

The validation of the first 212 household applications provides, for the first time, a solid empirical foundation for establishing realistic and meaningful targets for the rest of the programme period. When the original Outcome 2 targets were set, there was limited information on how households would respond to the loan facility, which technologies they would prefer, or how large the resulting impacts would be. Now that two full implementation years are complete, the observed results allow for a recalibration of expectations based on evidence rather than assumptions.

The revised targets presented in this chapter are built on two core components:

- projected number of financed households through 2027
- validated average impact per household, derived from the 212 completed projects.

The programme began with 30 financed households in 2024 and expanded rapidly to 182 in 2025. A moderate growth scenario for the coming years – 210 households in 2026 and 228 households in 2027 – would result in a total of 650 supported households over the 2024–2027 period. Applying the validated average impacts to this trajectory produces a new, evidence-based set of annual and cumulative targets, as shown in Table 10.

Table 10. Original Outcome 2 Targets vs. Updated Targets (2024–2027)

| Indicator                                   | Original Target | Updated Target (2024–2027) | Comment                                                   |
|---------------------------------------------|-----------------|----------------------------|-----------------------------------------------------------|
| <b>Energy savings (MWh/year)</b>            | 5,200           | <b>9,500</b>               | Higher due to scale and high heat pump uptake             |
| <b>CO<sub>2</sub> reductions (t/year)</b>   | 4,857           | <b>6,800</b>               | Based on a blended factor reflecting portfolio energy mix |
| <b>RES capacity (MW)</b>                    | 2.6             | <b>5.5</b>                 | Driven mainly by RES-eligible heat pumps                  |
| <b>PM<sub>2.5</sub> reduction (kg/year)</b> | –               | <b>7,500</b>               | Newly introduced target                                   |
| <b>PM<sub>10</sub> reduction (kg/year)</b>  | –               | <b>14,000</b>              | Newly introduced target                                   |

## Basis for updated calculations

### Average per-household impacts

Using the validated dataset from 212 households, the following average annual impacts were confirmed:

- **Energy savings:** 14.53 MWh/project
- **CO<sub>2</sub> emission reduction:** calculated using a weighted factor
- **RES capacity:** based on average intensity observed in 2025 (0.0105 MW/project)
- **PM reductions:**
  - PM<sub>2.5</sub>: 11.59 kg/project
  - PM<sub>10</sub>: 21.15 kg/project

These averages form the basis for projecting future outcomes, providing a representative benchmark for estimating expected impacts, as presented in Table 11.

Table 11. Updated Annual Targets – Outcome 2

| Year                   | Applications | Energy Savings (MWh/year) | CO <sub>2</sub> Reductions (t/year) | RES Capacity (MW/year) | PM <sub>2.5</sub> Reductions (kg/year) | PM <sub>10</sub> Reductions (kg/year) |
|------------------------|--------------|---------------------------|-------------------------------------|------------------------|----------------------------------------|---------------------------------------|
| <b>2024</b>            | 30           | 436                       | 309                                 | 0.14                   | 348                                    | 635                                   |
| <b>2025</b>            | 182          | 2,872                     | 2,039                               | 1.80                   | 2,202                                  | 3,863                                 |
| <b>2026</b>            | 210          | 3,000                     | 2,000                               | 1.5                    | 2,150                                  | 4,502                                 |
| <b>2027</b>            | 228          | 3,192                     | 2,452                               | 2.06                   | 2,800                                  | 5,000                                 |
| <b>Total 2024–2027</b> | <b>650</b>   | <b>9,500</b>              | <b>6,800</b>                        | <b>5.5</b>             | <b>7,500</b>                           | <b>14,000</b>                         |

Taken together, these updated targets present a much clearer picture of the programme's expected contribution by 2027. Energy savings more than double the original target, reflecting strong uptake and consistent performance of heat pump installations. CO<sub>2</sub> reductions follow this trend closely, reaching approximately 6,800 tons per year.

Renewable energy capacity grows substantially, reaching a cumulative 5.5 MW, largely due to the renewable classification of high-efficiency heat pumps. Although not originally targeted,

reductions in PM<sub>2.5</sub> and PM<sub>10</sub> emerge as some of the programme's most important co-benefits, with expected reductions of 7.5 tons of PM<sub>2.5</sub> and 14 tons of PM<sub>10</sub> annually by 2027.

These updated targets offer a realistic and evidence-based direction for the remainder of the programme period and form a strong basis for reporting, planning, and resource allocation.

## 2.8 Strategic Implications

The validation of the initial 212 projects and the definition of the 2027 targets highlight three distinct strategic imperatives for the remainder of the programme:

1. **Capitalizing on electrification:** The strong preference for heat pumps (47 per cent of installations) indicates that the market is moving naturally toward efficient electric heating. The programme should continue to streamline approval processes for this technology, as it is the primary driver of energy savings and CO<sub>2</sub> targets.
2. **Addressing the efficiency gap:** The limited uptake of envelope solutions (insulation and windows) represents a missed opportunity for deeper decarbonization. Without improving the thermal shell of buildings, the efficiency of the new heat pumps is not fully maximized. Future calls for applications should consider incentivizing "bundled" retrofits (e.g., Heat Pump + Insulation+PV).
3. **Strengthening methodological consistency and portfolio oversight:** The adoption of a harmonized blended emission factor (0.71 tCO<sub>2</sub>/MWh) improves the internal consistency between reported energy savings and aggregated CO<sub>2</sub> reductions. Going forward, continued monitoring of the portfolio's technology composition will support transparent reporting and help align climate mitigation outcomes with broader environmental co-benefits, including air quality improvements.

### 3. Contribution to improvement of the air quality and Sustainable Development Goals (SDGs)

#### 3.1 Contribution to improvement of the air quality

Air quality remains one of the most pressing environmental and public-health challenges in North Macedonia, particularly in urban areas and during the heating season. Elevated concentrations of particulate matter (PM<sub>10</sub> and PM<sub>2.5</sub>) are driven by a combination of factors, most notably residential heating based on solid fuels, traffic emissions, and industrial activities, compounded by unfavorable meteorological conditions and local topography. In this context, investments that promote energy efficiency and low-emission technologies are relevant not only for climate mitigation but also for improving local air quality.

Table 12 presents average annual concentrations for 2024 of PM<sub>2.5</sub> and PM<sub>10</sub> measured at selected monitoring stations operated by the Ministry of Environment and Physical Planning (MOEPP). The data indicate persistent air-quality pressures in several urban and industrial centers, including Skopje, Tetovo, Kichevo, Strumica, Kavadarci and Bitola.

In particular, PM<sub>10</sub> annual averages above or close to 40 µg/m<sup>3</sup> are observed at stations in Tetovo, Kičevo, Strumica, Kavadarci and parts of Skopje, while elevated PM<sub>2.5</sub> concentrations are recorded in several locations associated with dense residential areas and industrial activity. These patterns reflect the combined influence of household heating, traffic, and point and diffuse emissions from industrial sources, especially during winter months.

It should be noted that monitoring coverage is not uniform across the country, and several municipalities included in the programme do not have permanent monitoring stations. Nevertheless, the 2024 data provide a representative snapshot of spatial air-pollution pressures and forms an appropriate baseline for assessing the relevance of programme interventions.

*Table 12. Average annual PM<sub>2.5</sub> and PM<sub>10</sub> concentrations at selected air-quality monitoring stations (2024)<sup>4</sup>*

| Station         | PM <sub>2.5</sub> Average annual concentration (µg/m <sup>3</sup> ) | PM <sub>10</sub> Average annual concentration (µg/m <sup>3</sup> ) |
|-----------------|---------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Berovo</b>   | -                                                                   | -                                                                  |
| <b>Bitola 1</b> | 20.43                                                               | -                                                                  |
| <b>Bitola 2</b> | 27.33                                                               | 40.22                                                              |
| <b>Veles 2</b>  | 19.98                                                               | 42.08                                                              |

<sup>4</sup> Source: <https://air.moepp.gov.mk/Pages/DataAnalysis>



|                              |       |       |
|------------------------------|-------|-------|
| <b>Gazi Baba</b>             | 22.22 | 32.77 |
| <b>Gevgelija</b>             | -     | -     |
| <b>Gostivar</b>              | 26.84 | 39.89 |
| <b>Kavadarci</b>             | 26.03 | 46.89 |
| <b>Karposh</b>               | -     | 35.69 |
| <b>Kichevo</b>               | 32.84 | 49.88 |
| <b>Kochani</b>               | 15.07 | -     |
| <b>Kumanovo</b>              | -     | -     |
| <b>Lazaropole</b>            | -     | 11.72 |
| <b>Lisiche</b>               | 28.95 | 49.52 |
| <b>Miladinovci</b>           | 17.60 | 27.21 |
| <b>Mobile Gjorche Petrov</b> | 24.18 | 31.07 |
| <b>Ohrid</b>                 | 20.96 | 34.53 |
| <b>Prilep</b>                | -     | -     |
| <b>Rektorat</b>              | -     | 41.40 |
| <b>Strumica</b>              | 29.19 | 49.69 |
| <b>Tetovo</b>                | 30.07 | 43.25 |
| <b>Centar</b>                | -     | 43.91 |

To support visual interpretation, the locations of the air quality monitoring stations are indicated on the map in Figure 7. The country is overlaid with a regular grid, and grid cells are colour-coded to reflect the range of measured particulate-matter concentrations at nearby monitoring stations. This representation is intended solely to provide a simplified visual overview of air quality conditions and their spatial context. It does not constitute a dispersion model or spatial interpolation of pollutant concentrations, nor does it imply uniform air quality conditions within individual grid cells. The map therefore serves as an illustrative tool to contextualize the distribution of applications in relation to areas with known air quality pressures.

### **Spatial distribution of applications – Outcome 1 (SMEs)**

The geographical distribution of SME applications under Outcome 1 shows a strong concentration in Skopje, which accounts for the largest share of applications. This reflects the higher density of SMEs, stronger investment capacity, and greater awareness of available financing instruments in the capital. Beyond Skopje, notable uptake is observed in Strumica, Bitola, Tetovo and Kavadarci, all of which are important regional economic and industrial centers with a strong manufacturing, agri-processing, and service base.

Several of these cities – particularly Skopje, Tetovo, Bitola and Kavadarci – are also locations where industrial activity contributes to local air pollution pressures, alongside residential heating and transport. The interest shown by SMEs in these areas is therefore particularly relevant, as

investments in energy efficiency, cleaner production processes and modern heating systems can contribute to reducing both energy consumption and local emissions from industrial and commercial sources.

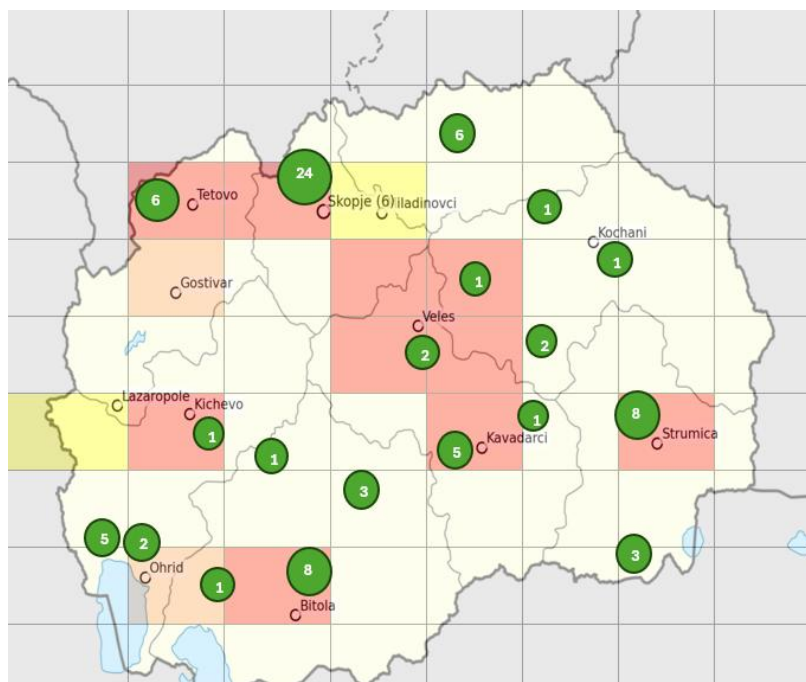
Municipalities with lower participation tend to be smaller or more rural, suggesting that additional outreach and tailored support may be required to further expand SME engagement across the country.

Table 13 presents the geographical distribution of application in Outcome 1.

*Table 13. Geographical distribution of applications – Outcome 1*

| Location     | Applications |
|--------------|--------------|
| Bitola       | 8            |
| Gevgelija    | 3            |
| Kavadarci    | 5            |
| kochani      | 1            |
| Kichevo      | 1            |
| Kumanovo     | 6            |
| Negotino     | 2            |
| Ohrid        | 2            |
| Plasnica     | 1            |
| Prilep       | 3            |
| Probishtip   | 1            |
| Resen        | 1            |
| Skopje       | 24           |
| Shtip        | 2            |
| Struga       | 5            |
| Strumica     | 8            |
| Sveti Nikole | 1            |
| Tetovo       | 6            |
| Veles        | 2            |

Figure 7 illustrates the spatial distribution of SME applications. The map uses a uniform grid-based representation to aggregate applications spatially and to highlight general regional patterns rather than precise point locations. The chosen resolution supports comparability across regions while avoiding over-interpretation at the municipal or facility level.



## Legend

- Number of household applications
- Low PM concentration ■ Moderate PM concentration ■ High PM concentration
- Grid: air-quality interpolation (non-administrative)

Figure 7. Geographical distribution of applications – Outcome 1<sup>5</sup>

## Spatial distribution of applications – Outcome 2 (Households)

Under Outcome 2, the spatial distribution of household applications shows an even clearer concentration in larger urban centers. The highest number of applications is recorded in Skopje, followed by Prilep, Ohrid, Strumica, Tetovo and Struga. This distribution broadly reflects population density, housing stock characteristics, and access to information and financial instruments.

Many of these cities also experience significant air-quality challenges, where residential heating, traffic and nearby industrial activities jointly contribute to elevated particulate-matter concentrations. Skopje and Tetovo, in particular, are well known for recurring winter smog episodes dominated by PM<sub>10</sub> and PM<sub>2.5</sub>. In this context, the replacement of inefficient heating

<sup>5</sup> Image source: Source: Wikimedia Commons, *North Macedonia location map*, CC BY-SA 3.0. Modified by the author.

systems with energy-efficient and low-emission technologies—such as heat pumps and modern biomass systems—has particular relevance for reducing emissions at the household level.

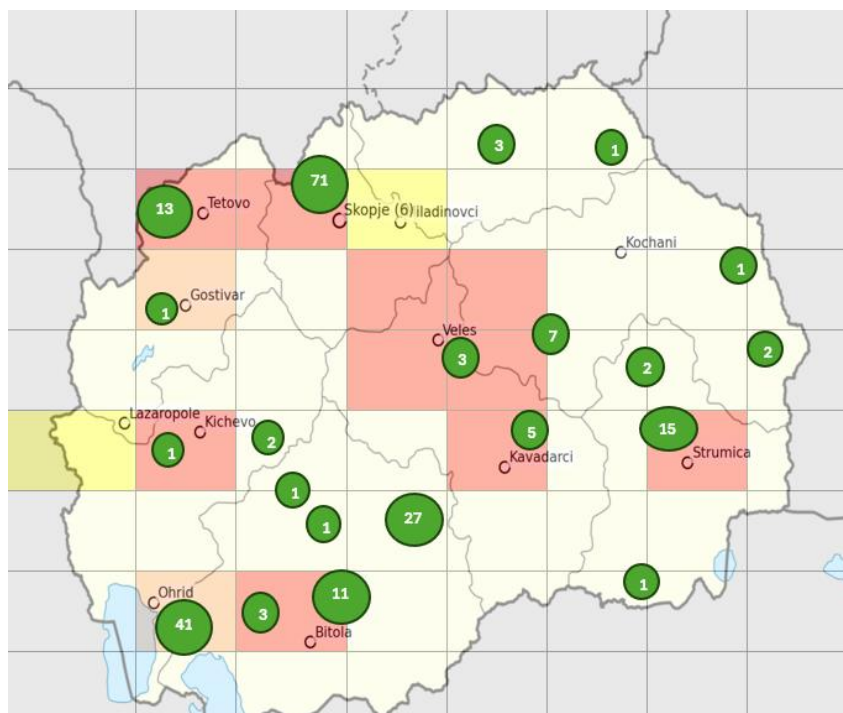
Municipalities with fewer applications should not be interpreted as areas with lower air-pollution pressures, but rather as locations with lower population density or where additional awareness-raising and facilitation may be required to increase participation.

Table 14 presents the geographical distribution of application in Outcome 2.

*Table 14. Geographical distribution of applications – Outcome 2*

| Location        | Applications |
|-----------------|--------------|
| Berovo          | 2            |
| Bitola          | 11           |
| Delchevo        | 1            |
| Gevgelija       | 1            |
| Gostivar        | 1            |
| Kavadarci       | 5            |
| Kichevo         | 1            |
| Kriva Palanka   | 1            |
| Krivogashtani   | 1            |
| Krushevo        | 1            |
| Kumanovo        | 3            |
| Makedonski brod | 1            |
| Ohrid           | 27           |
| Plasnica        | 1            |
| Prilep          | 27           |
| Radovish        | 2            |
| Resen           | 3            |
| Shtip           | 7            |
| Skopje          | 71           |
| Struga          | 14           |
| Strumica        | 15           |
| Tetovo          | 13           |
| Veles           | 3            |

Figure 8 presents the geographical distribution of household applications using the same grid-based mapping approach as for Outcome 1, ensuring consistency and comparability across outcomes.



## Legend

● Number of household applications

■ Low PM concentration

■ Moderate PM concentration

■ High PM concentration

□ Grid: air-quality interpolation (non-administrative)

Figure 8. Geographical distribution of applications – Outcome 2<sup>6</sup>

## Contribution to air quality improvement

While changes in ambient air-pollution concentrations cannot be directly attributed to individual programme interventions, the aggregated impact of supported investments is clearly aligned with national and local air-quality objectives. By promoting energy efficiency, fuel switching and low-emission technologies in areas where air-pollution pressures are influenced by residential heating, traffic and industrial activity, the programme contributes to reducing emissions at their sources.

The spatial overlap between higher application density and urban and industrial centers with monitored air-quality challenges strengthens the relevance of the programme from an air-quality perspective. In combination with broader regulatory, municipal and sectoral solutions, the supported investments under Outcomes 1 and 2 represent a meaningful contribution towards

<sup>6</sup> Image source: Source: Wikimedia Commons, North Macedonia location map, CC BY-SA 3.0. Modified by the author.

reducing particulate-matter emissions, improving living conditions, and supporting public-health objectives, particularly in areas most affected by winter pollution episodes.

### 3.2 Contribution to Sustainable Development Goals (SDGs)

The updated performance metrics confirm that the Green Finance Facility To Improve Air Quality And Combat Climate Change In North Macedonia Outcome 1 and 2 directly accelerates North Macedonia's progress toward the specific SDG targets:

- **SDG 7 (Energy Efficiency):** The validated average savings of 68,653 MWh exceeds initial planning assumptions, directly contributing to the national target of reducing final energy consumption.
- **SDG 7 (Renewable Energy):** By categorizing high-efficiency heat pumps as renewable energy (in line with EU Directive 2018/2001), the programme has added 44 MW of RES capacity to date.
- **SDG 13 (Climate Action):** The revised monitoring framework ensures that the programme's contribution to national GHG inventories is transparent and verifiable, supporting the Nationally Determined Contribution (NDC) target of 51 per cent emissions reduction. The programme is directly responsible for 53,050 tCO<sub>2</sub> reduction.
- **SDG 3 (Health/Air Pollution):** Although not originally a primary indicator, the quantification of PM<sub>2.5</sub> and PM<sub>10</sub> reductions (7,592 kg/yr) demonstrates a critical co-benefit for public health in urban areas, directly addressing the pollution issues cited in the baseline analysis.

Figure 9 illustrates the contribution of the programme to selected Sustainable Development Goals, highlighting impacts on clean energy access, climate change mitigation, and air-quality-related health outcomes.

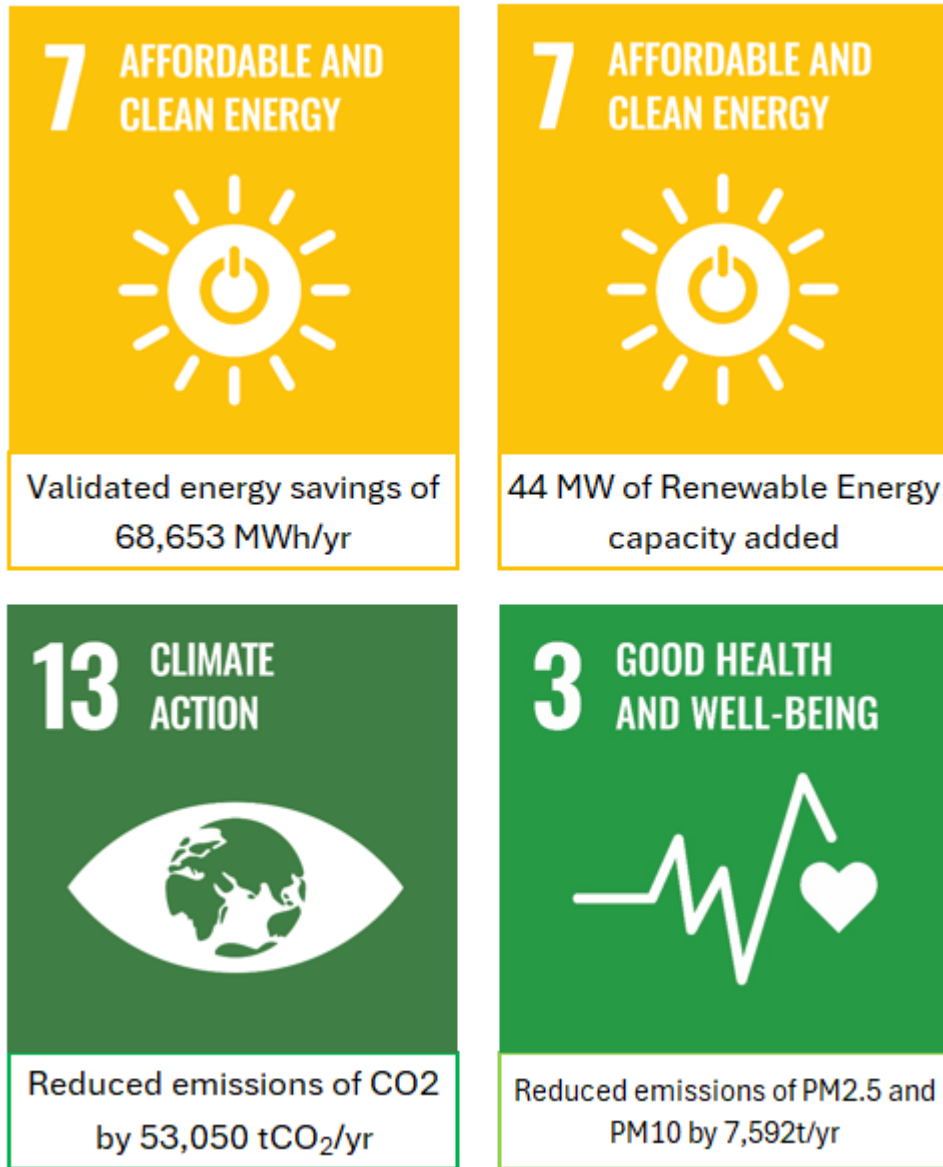


Figure 9. Contribution to the Sustainable Development Goals<sup>7</sup>

<sup>7</sup> Source: SDG icons: United Nations Sustainable Development Goals. SDG icons are reproduced in accordance with the United Nations Sustainable Development Goals guidelines



## 4. Sensitivity

### Outcome 1

#### 4.1 Overview and Risk Scenarios

Forecasting programme performance for industrial investments through 2027 involves inherent uncertainties related to investment decisions, technology preferences, and operational conditions at enterprise level. To ensure the robustness of the updated portfolio-level targets, a sensitivity analysis was conducted to stress-test the programme against key risks identified in the Uncertainty Management Strategy (Section 2.5).

Rather than relying on a single projection, this analysis defines a performance floor and a target range by quantifying the potential impact of the following scenarios:

- **Scenario A: Investment pipeline risk (Volume risk)** - This scenario assumes a slowdown in industrial investment decisions due to macroeconomic uncertainty, delayed capital expenditure, or more cautious corporate planning. It models a 25 per cent reduction in the number of approved projects compared to the base case.
  - *Primary Impact:* A proportional reduction across all key indicators, including Energy Savings, CO<sub>2</sub> Reductions, RES Capacity, and PM reductions, reflecting lower overall implementation volume.
- **Scenario B: Technology mix shift (Structure risk)** - This scenario assumes a shift in the composition of approved projects toward energy efficiency-focused interventions, driven by factors such as grid connection constraints, permitting delays for generation assets, or changes in electricity price expectations. The total number of projects remains unchanged, but the share of planned RES and EE+RES projects is replaced by stand-alone EE solutions.
  - *Primary Impact:* Energy savings remain broadly stable or decline only moderately, as EE solutions typically deliver high savings per unit of investment; RES capacity is significantly reduced, reflecting the lower share of generation-based projects; CO<sub>2</sub> reductions decline moderately, as a larger share of savings displaces electricity rather than higher-emission fuels; PM reductions remain stable or improve, due to reduced reliance on combustion-based technologies.
- **Scenario C: Performance realization risk (Operational risk)** - This scenario reflects the risk that industrial systems do not achieve their full expected performance due to commissioning delays, operational constraints, or sub-optimal integration into existing

processes. It models a 15 per cent reduction in realized energy savings relative to design expectations.

- *Primary Impact: A reduction in Energy Savings and CO<sub>2</sub> Reductions, while RES capacity and the number of beneficiaries remain unaffected.*
- **Scenario D: Combined conservative scenario (Minimum performance floor)** - This scenario represents a “perfect storm” in which reduced investment volume, a shift toward lower-capacity RES deployment, and lower-than-expected operational performance occur simultaneously. It defines the minimum compliance floor for Outcome 1. Performance below this level would indicate structural constraints rather than normal implementation variability.
- **Scenario E: Upside case (Optimistic)** - This scenario assumes favorable market conditions, including stronger investment confidence, faster approval cycles, and larger-than-average project scale. It defines the optimistic ceiling of what the industrial portfolio could achieve under accelerated implementation and high system performance.

The quantitative impacts of each scenario are assessed relative to the base case targets established in Chapter Three and are presented in Tables 15-19.

#### Scenario A – Investment pipeline risk (Volume risk)

Table 15. Scenario A – Investment Pipeline Risk (Volume Risk) - Outcome 1

| Indicator                               | Base    | Scenario A (–<br>25% applications) | Change  |
|-----------------------------------------|---------|------------------------------------|---------|
| <b>Energy savings (MWh/yr)</b>          | 104,000 | <b>78,000</b>                      | -26,000 |
| <b>CO<sub>2</sub> reductions (t/yr)</b> | 95,000  | <b>71,250</b>                      | -23,750 |
| <b>RES capacity (MW)</b>                | 70      | <b>52.50</b>                       | -17.50  |
| <b>PM<sub>2.5</sub> (g/MWh/yr)</b>      | 1,103   | <b>827</b>                         | -276    |
| <b>PM<sub>10</sub> (g/MWh/yr)</b>       | 2,054   | <b>1,541</b>                       | -514    |

## Scenario B: Technology mix shift (Structure risk)

Table 16. Scenario B: Technology Mix Shift (Structure Risk) - Outcome 1

| Indicator                         | Base    | Scenario B    | Change |
|-----------------------------------|---------|---------------|--------|
| Energy savings (MWh/yr)           | 104,000 | <b>99,750</b> | -4,250 |
| CO <sub>2</sub> reductions (t/yr) | 95,000  | <b>85,500</b> | -9,500 |
| RES capacity (MW)                 | 70.0    | <b>49.00</b>  | -21.00 |
| PM <sub>2.5</sub> (g/MWh/yr)      | 1,103   | <b>1,103</b>  | -      |
| PM <sub>10</sub> (g/MWh/yr)       | 2,054   | <b>2,054</b>  | -      |

## Scenario C: Performance realization risk (Operational risk)

Table 17. Scenario C: Performance Realization Risk (Operational Risk) - Outcome 1

| Indicator                         | Base    | Scenario C    | Change  |
|-----------------------------------|---------|---------------|---------|
| Energy savings (MWh/yr)           | 104,000 | <b>88,400</b> | -15,600 |
| CO <sub>2</sub> reductions (t/yr) | 95,000  | <b>80,750</b> | -14,250 |
| RES capacity (MW)                 | 70.0    | <b>5.50</b>   | -65     |
| PM <sub>2.5</sub> (g/MWh/yr)      | 1,103   | <b>938</b>    | -165    |
| PM <sub>10</sub> (g/MWh/yr)       | 2,054   | <b>1,746</b>  | -308    |

## Scenario D: Combined conservative scenario (Minimum performance floor)

Table 18. Scenario D: Combined Conservative Scenario (Minimum Performance Floor) - Outcome 1

| Indicator                         | Base    | Scenario D (Updated) | Change  |
|-----------------------------------|---------|----------------------|---------|
| Energy savings (MWh/yr)           | 104,000 | <b>54,366</b>        | -49,634 |
| CO <sub>2</sub> reductions (t/yr) | 95,000  | <b>49,661</b>        | -45,339 |
| RES capacity (MW)                 | 70.0    | <b>43.05</b>         | -26.95  |
| PM <sub>2.5</sub> (g/MWh/yr)      | 1,103   | <b>577</b>           | -526    |
| PM <sub>10</sub> (g/MWh/yr)       | 2,054   | <b>1,074</b>         | -980    |

## Scenario E: Upside case (Optimistic)

Table 19. Scenario E: Upside Case (Optimistic) - Outcome 1

| Indicator                         | Base    | Scenario E     | Change  |
|-----------------------------------|---------|----------------|---------|
| Energy savings (MWh/yr)           | 104,000 | <b>130,000</b> | +26,000 |
| CO <sub>2</sub> reductions (t/yr) | 95,000  | <b>118,750</b> | +23,750 |
| RES capacity (MW)                 | 70.0    | <b>87.50</b>   | +17.50  |
| PM <sub>2.5</sub> (g/MWh/yr)      | 1,103   | <b>1,379</b>   | +276    |
| PM <sub>10</sub> (g/MWh/yr)       | 2,054   | <b>2,568</b>   | +514    |

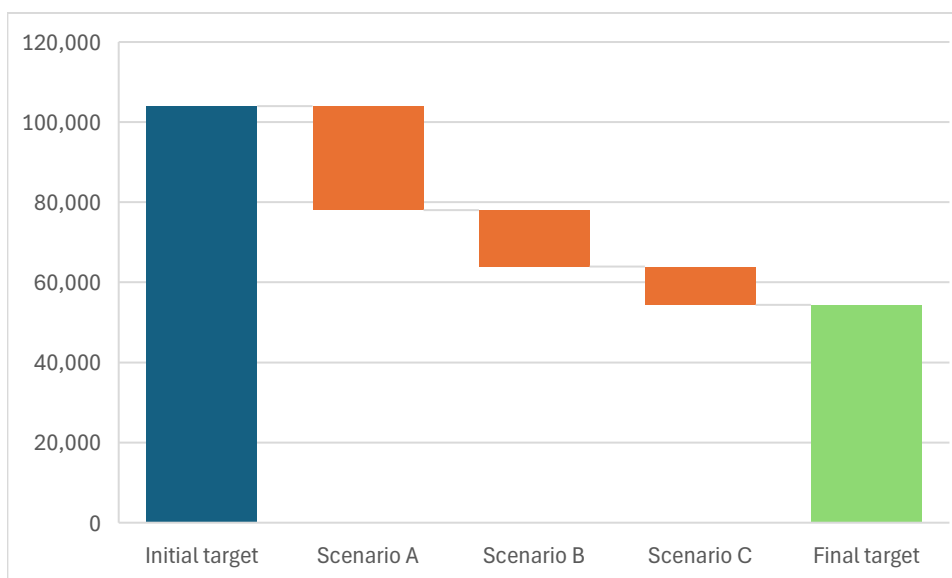


Figure 10. Energy savings under Scenario D – Outcome 1

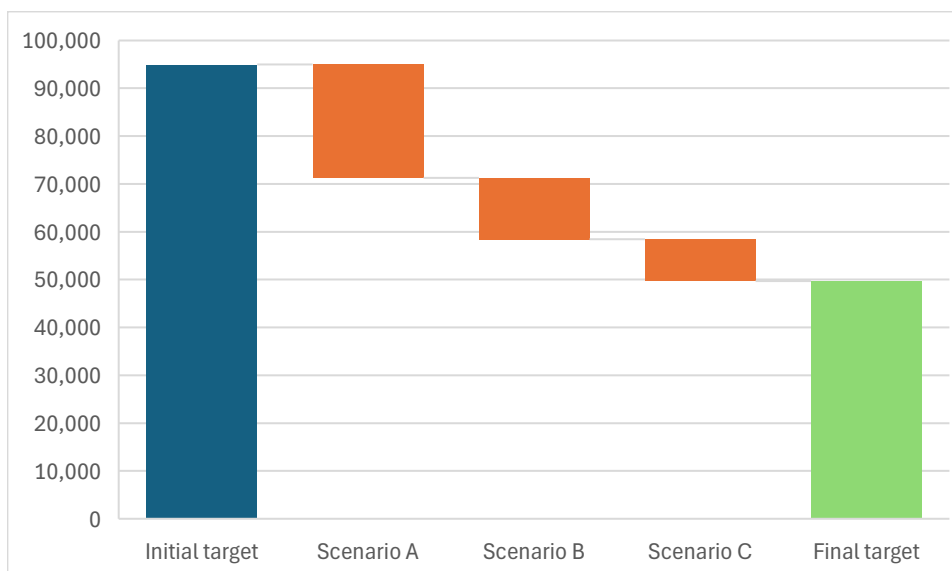


Figure 11. CO<sub>2</sub> reductions under Scenario D – Outcome 1

Figure 10 and Figure 11 show the impact of successive adjustment scenarios on projected energy savings and CO<sub>2</sub> reductions, resulting in the final target under Scenario D for Outcome 1.

## 4.2 Summary Comparison

As presented in Table 20 and Figure 12, the analysis confirms that while Volume Risk (Scenario A) has the largest impact, the Technology Shift (Scenario B) does not pose a significant risk. The

programme's performance should be monitored against the range defined by Scenario D (Minimum) and Scenario E (Optimistic).

Table 20. Summary Comparison – Outcome 1

| Scenario               | Energy Savings (MWh/yr) | CO <sub>2</sub> Reductions (t/yr) | RES Capacity (MW) | PM <sub>2.5</sub> (g/MWh/yr) | PM <sub>10</sub> (g/MWh/yr) |
|------------------------|-------------------------|-----------------------------------|-------------------|------------------------------|-----------------------------|
| Updated Targets (Base) | <b>104,000</b>          | <b>95,000</b>                     | <b>70.0</b>       | <b>1,103</b>                 | <b>2,054</b>                |
| Scenario A             | <b>78,000</b>           | <b>71,250</b>                     | <b>52.50</b>      | <b>827</b>                   | <b>1,541</b>                |
| Scenario B             | <b>99,750</b>           | <b>85,500</b>                     | <b>49.00</b>      | <b>1,103</b>                 | <b>2,054</b>                |
| Scenario C             | <b>88,400</b>           | <b>80,750</b>                     | <b>5.50</b>       | <b>938</b>                   | <b>1,746</b>                |
| Scenario D             | <b>54,366</b>           | <b>49,661</b>                     | <b>43.05</b>      | <b>577</b>                   | <b>1,074</b>                |
| Scenario E             | <b>130,000</b>          | <b>118,750</b>                    | <b>87.50</b>      | <b>1,379</b>                 | <b>2,568</b>                |

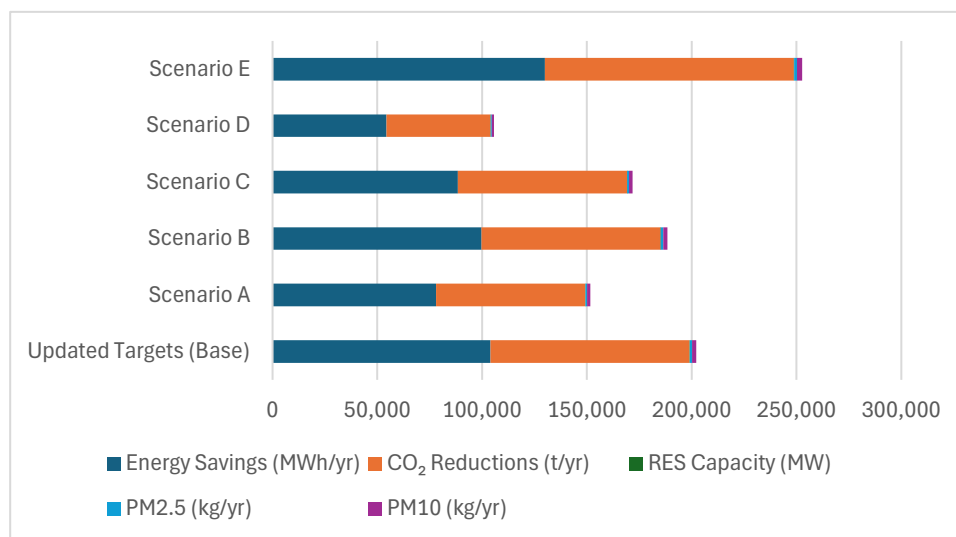


Figure 12. Summary Comparison – Outcome 1

## Outcome 2

### 4.3 Overview and Risk Scenarios

Forecasting programme performance through 2027 carries inherent uncertainties related to market behavior, technology trends, and user operations. To ensure the robustness of the updated targets, a sensitivity analysis was conducted to stress-test the programme against specific risk scenarios defined in the Uncertainty Management Strategy (Section 2.5).

Rather than relying on a single projection, this analysis establishes a "performance floor" and a "target range" by quantifying the impact of the following scenarios:

- **Scenario A: Market saturation (Volume risk)** This scenario assumes a slowdown in demand as the programme moves beyond "early adopters" to harder-to-reach market segments. It models a 25 per cent reduction in total applications compared to the base growth forecast.
  - *Primary Impact:* Proportional reduction across all indicators (Energy, CO<sub>2</sub>, RES).
- **Scenario B: Technology shift (Mix risk)** This scenario assumes a shift in user preference away from electrification due to potential electricity price hikes or grid constraints. It models a 25 per cent reduction in Heat Pump installations, replaced largely by biomass solutions.
  - *Primary Impact:* Moderate reduction in Energy Savings (due to lower efficiency of biomass vs. heat pumps) but a severe reduction in Air Quality benefits, as biomass boilers emit significantly more Particulate Matter (PM) than heat pumps.
- **Scenario C: Rebound effect (Performance risk)** This scenario accounts for behavioural "take-back," where households utilize the financial savings from energy efficiency to increase indoor comfort (e.g., raising thermostat settings) rather than reducing consumption. It models a 15 per cent drop in realized savings across all technologies.
  - *Primary Impact:* Reduction in Energy and CO<sub>2</sub> savings without affecting the number of beneficiaries or RES capacity.
- **Scenario D: Combined conservative (Minimum compliance floor)** This represents a "perfect storm" scenario where market saturation, technology shifts, and rebound effects occur simultaneously. It serves as the absolute minimum performance floor for the programme. If the programme falls below this level, it indicates structural failure rather than variance.

- **Scenario E: Upside case (Optimistic)** This scenario assumes favorable market conditions, including accelerated adoption rates and higher-than-average system performance. It defines the "Optimistic Ceiling" of what the programme could achieve under ideal conditions.

Tables 21-25 present the quantitative impact of each scenario compared to the "Base Case" targets established in Chapter Three.

#### Scenario A – 25 per cent less applications (Market volume risk)

Table 21. Scenario A – 25 per cent less applications (Market Volume Risk) – Outcome 2

| Indicator                         | Base   | Scenario A (–25%) | Change |
|-----------------------------------|--------|-------------------|--------|
| Energy savings (MWh/yr)           | 9,500  | <b>7,125</b>      | -2,375 |
| CO <sub>2</sub> reductions (t/yr) | 6,800  | <b>5,100</b>      | -1,700 |
| RES capacity (MW)                 | 5.5    | <b>4.13</b>       | -1.38  |
| PM <sub>2.5</sub> (kg/yr)         | 7,500  | <b>5,625</b>      | -1,875 |
| PM <sub>10</sub> (kg/yr)          | 14,000 | <b>10,500</b>     | -3,500 |

#### Scenario B – 25 per cent less heat pumps (Technology mix risk)

Table 22. Scenario B – 25 per cent less heat pumps (Technology Mix Risk) – Outcome 2

| Indicator                         | Base   | Scenario B    | Change |
|-----------------------------------|--------|---------------|--------|
| Energy savings (MWh/yr)           | 9,500  | <b>7,790</b>  | -1,710 |
| CO <sub>2</sub> reductions (t/yr) | 6,800  | <b>5,576</b>  | -1,224 |
| RES capacity (MW)                 | 5.5    | <b>4.51</b>   | -0.99  |
| PM <sub>2.5</sub> (kg/yr)         | 7,500  | <b>6,150</b>  | -1,350 |
| PM <sub>10</sub> (kg/yr)          | 14,000 | <b>11,480</b> | -2,520 |

#### Scenario C – Reduced system performance influence (Rebound effect)

Table 23. Scenario C – Reduced system performance influence – (Rebound Effect) – Outcome 2

| Indicator | Base | Scenario C | Change |
|-----------|------|------------|--------|
|-----------|------|------------|--------|



|                                         |        |               |        |
|-----------------------------------------|--------|---------------|--------|
| <b>Energy savings (MWh/yr)</b>          | 9,500  | <b>8,075</b>  | -1,425 |
| <b>CO<sub>2</sub> reductions (t/yr)</b> | 6,800  | <b>5,780</b>  | -1,020 |
| <b>RES capacity (MW)</b>                | 5.5    | <b>5.50</b>   | 0      |
| <b>PM<sub>2.5</sub> (kg/yr)</b>         | 7,500  | <b>6,375</b>  | -1,125 |
| <b>PM<sub>10</sub> (kg/yr)</b>          | 14,000 | <b>11,900</b> | -2,100 |

#### **Scenario D – Combined conservative case (Minimum compliance floor)**

This scenario establishes the minimum floor for programme performance. Even under these adverse conditions, the programme is expected to deliver 5,000 MWh of savings annually.

*Table 24. Scenario D – Combined Conservative Case – Outcome 2*

| <b>Indicator</b>                           | <b>Base</b> | <b>Scenario D</b> | <b>Change</b> |
|--------------------------------------------|-------------|-------------------|---------------|
| <b>Energy savings (MWh/yr)</b>             | 9,500       | <b>4,966</b>      | -4,534        |
| <b>CO<sub>2</sub> reductions (t/yr)</b>    | 6,800       | <b>3,555</b>      | -3,245        |
| <b>RES capacity (MW)</b>                   | 5.5         | <b>3.38</b>       | -2.12         |
| <b>PM<sub>2.5</sub> reductions (kg/yr)</b> | 7,500       | <b>3,921</b>      | -3,579        |
| <b>PM<sub>10</sub> reductions (kg/yr)</b>  | 14,000      | <b>7,319</b>      | -6,682        |

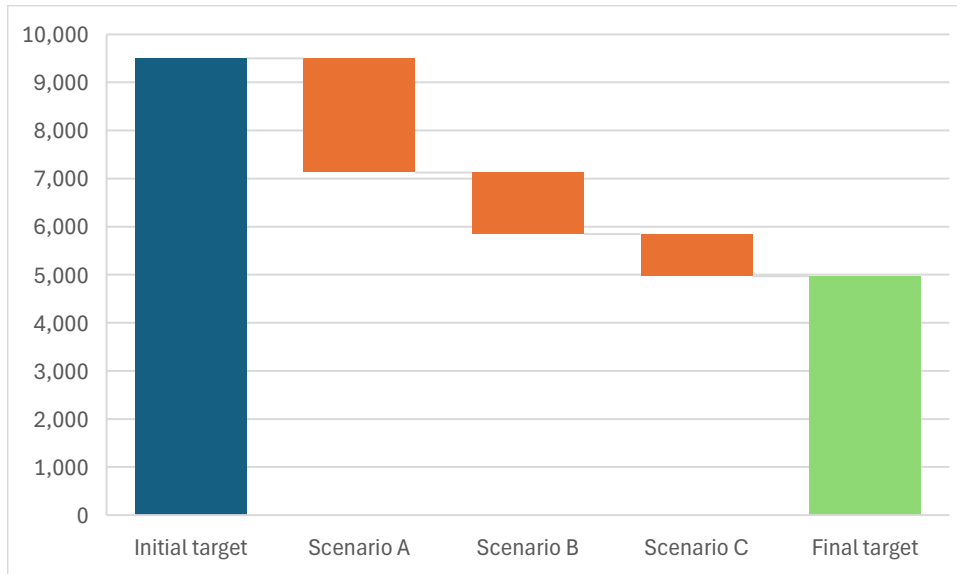


Figure 13. Energy savings under Scenario D – Outcome 2



Figure 14. CO<sub>2</sub> reductions under Scenario D – Outcome 2

Figure 13 and Figure 14 show the impact of successive adjustment scenarios on projected energy savings and CO<sub>2</sub> reductions, resulting in the final target under Scenario D for Outcome 2.

### Scenario E – Upside case

Table 25. Scenario E – Upside Case – Outcome 2

| Indicator                               | Base   | Scenario E    | Change |
|-----------------------------------------|--------|---------------|--------|
| <b>Energy savings (MWh/yr)</b>          | 9,500  | <b>11,875</b> | +2,375 |
| <b>CO<sub>2</sub> reductions (t/yr)</b> | 6,800  | <b>8,500</b>  | +1,700 |
| <b>RES capacity (MW)</b>                | 5.5    | <b>6.88</b>   | +1.38  |
| <b>PM<sub>2.5</sub> (kg/yr)</b>         | 7,500  | <b>9,375</b>  | +1,875 |
| <b>PM<sub>10</sub> (kg/yr)</b>          | 14,000 | <b>17,501</b> | +3,500 |

#### 4.4 Summary Comparison

As presented in Table 26, Figure 15, and Figure 16, the analysis confirms that while Volume Risk (Scenario A) has the largest impact, the Technology Shift (Scenario B) presents a specific structural risk to RES and Air Quality targets. The programme's performance should be monitored against the range defined by Scenario D (Minimum) and Scenario E (Optimistic).

Table 26. Summary Comparison – Outcome 2

| Scenario               | Energy Savings (MWh/yr) | CO <sub>2</sub> Reductions (t/yr) | RES Capacity (MW) | PM <sub>2.5</sub> (kg/yr) | PM <sub>10</sub> (kg/yr) |
|------------------------|-------------------------|-----------------------------------|-------------------|---------------------------|--------------------------|
| Updated Targets (Base) | <b>9,500</b>            | <b>6,800</b>                      | <b>5.5</b>        | <b>7,500</b>              | <b>14,000</b>            |
| Scenario A             | <b>7,125</b>            | <b>5,100</b>                      | <b>4.13</b>       | <b>5,625</b>              | <b>10,500</b>            |
| Scenario B             | <b>7,790</b>            | <b>5,576</b>                      | <b>4.51</b>       | <b>6,150</b>              | <b>11,480</b>            |
| Scenario C             | <b>8,075</b>            | <b>5,780</b>                      | <b>5.50</b>       | <b>6,375</b>              | <b>11,900</b>            |
| Scenario D             | <b>4,966</b>            | <b>3,555</b>                      | <b>3.38</b>       | <b>3,921</b>              | <b>7,319</b>             |
| Scenario E             | <b>11,875</b>           | <b>8,500</b>                      | <b>6.88</b>       | <b>9,375</b>              | <b>17,501</b>            |

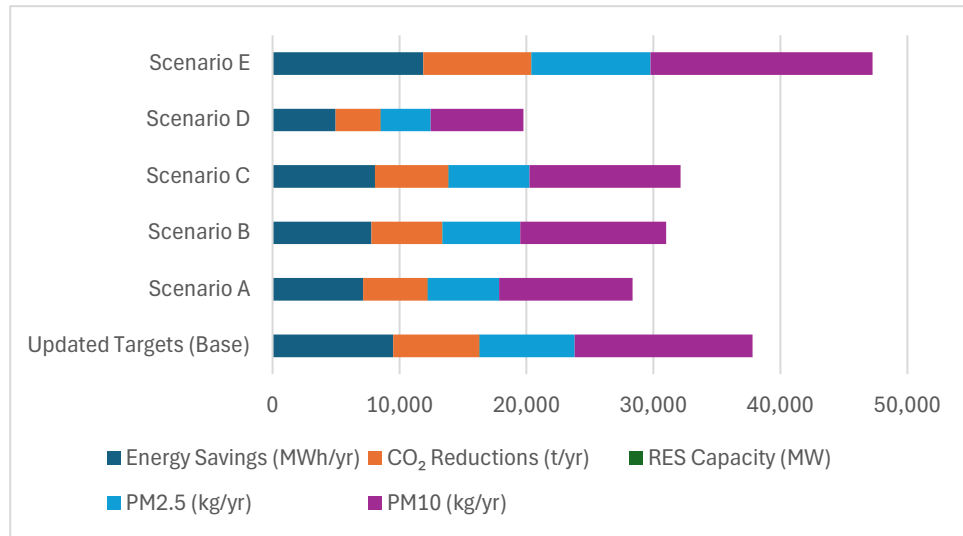


Figure 15. Summary Comparison – Outcome 2

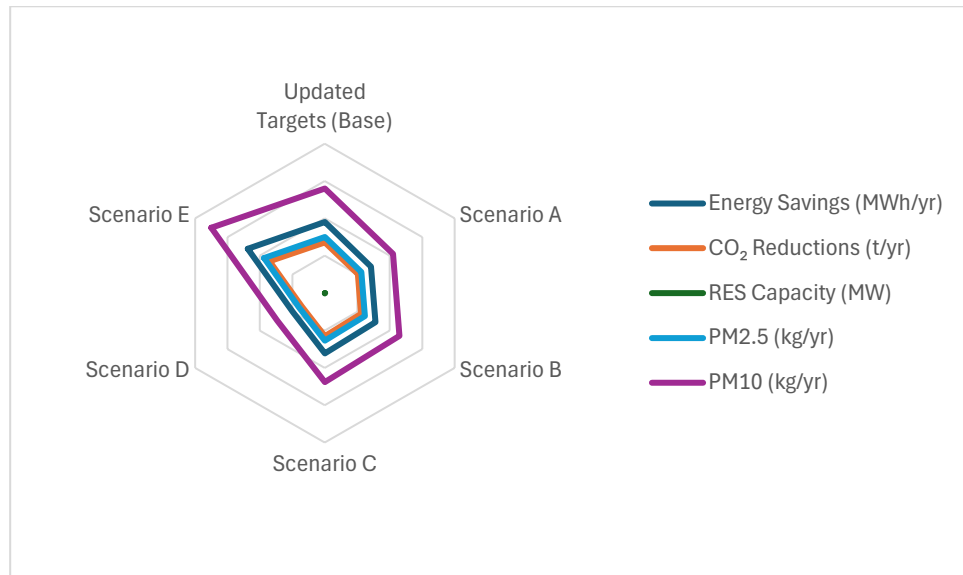


Figure 16. Radar preview of selected scenarios – Outcome 2

## 5. Standardized Methodology for Energy and Emission Valuation

This methodology defines a standardized approach for calculating energy savings and greenhouse gas (GHG) emission reductions at programme level. It ensures consistency and comparability of results across technologies, households, and reporting periods.

The methodology harmonizes existing consultant calculations and addresses inconsistencies observed in earlier reporting cycles, particularly in the treatment of heating systems and emission valuation.

### **System boundary and baseline definition**

The calculation boundary is defined at the level of the residential unit benefiting from programme support. Only operational energy consumption related to space heating and domestic hot water is considered.

Baseline energy demand is calculated using a standardized method assuming full-dwelling heating, in accordance with consultant calculation procedures and applicable national rulebooks. This standardization supports comparability across the programme portfolio.

It is recognized that some households may have heated only part of the dwelling prior to the intervention. In such cases, installation of a new or centralized heating system may lead to increased comfort and expanded heated area. Reported energy savings therefore represent reductions relative to a standardized service level, rather than guaranteed reductions in metered household energy consumption.

### **Energy accounting approach**

To ensure physical consistency, baseline and post-intervention energy use are assessed on the same energy basis.

For combustion-based heating systems, final energy input (fuel) is converted into useful heat using system efficiency ( $\eta$ ). For electrically driven heat pumps, the final energy, electricity input is linked to useful heat delivery using a seasonal performance factor (SPF or seasonal COP), reflecting the contribution of renewable ambient heat.

This approach avoids mixing useful and final energy concepts within the same calculation and ensures that energy savings and emission reductions are derived from equivalent heating services.

Where system-specific data are unavailable, standard default values are applied.

Table 27 summarizes the default efficiency and performance parameters assumed for existing heating systems, as defined in the national Rulebook on Energy Audits, and applied consistently across the portfolio where system-specific data are not available.

Table 27. Default Efficiency and Performance Parameters of Existing Systems

| Heating system type                | Parameter | Default value | Source            |
|------------------------------------|-----------|---------------|-------------------|
| <b>Liquid fuel boiler</b>          | $\eta$    | 0.75-0.85     | National rulebook |
| <b>Solid fuel boiler</b>           | $\eta$    | 0.65-0.80     | National rulebook |
| <b>Electric resistance heating</b> | $\eta$    | 0.997         | National rulebook |

### Emission calculation

GHG emissions are calculated by applying standardized CO<sub>2</sub> emission factors to final energy consumption before and after the intervention. Emission reductions are calculated as the difference between baseline and post-intervention emissions.

Emission factors are applied consistently across the programme and aligned with nationally approved methodologies.

Table 28 presents the CO<sub>2</sub> emission factors for different energy carriers, expressed in both tCO<sub>2</sub>/TJ and tCO<sub>2</sub>/MWh, as defined in the national Rulebook on Energy Audits.

Table 28. CO<sub>2</sub> Emission Factors<sup>8</sup>

| Energy carrier                 | Emission factor (tCO <sub>2</sub> /TJ) | Emission factor (tCO <sub>2</sub> /MWh) | Source            |
|--------------------------------|----------------------------------------|-----------------------------------------|-------------------|
| <b>Extra light heating oil</b> | 74.1                                   | 0.267                                   | National rulebook |
| <b>Natural gas</b>             | 56.1                                   | 0.202                                   | National rulebook |
| <b>Heavy fuel oil</b>          | 77.4                                   | 0.279                                   | National rulebook |
| <b>Firewood</b>                | 0/112                                  | 0/0.403*                                | National rulebook |
| <b>Lignite (brown coal)</b>    | 101.2                                  | 0.364                                   | National rulebook |

<sup>8</sup> Rulebook on energy audits (Official Gazette of the Republic of North Macedonia, No. 94/2013, 4 July 2013)

|                             |       |       |                   |
|-----------------------------|-------|-------|-------------------|
| <b>Petrol (gasoline)</b>    | 69.3  | 0.249 | National rulebook |
| <b>Diesel fuel</b>          | 74.1  | 0.267 | National rulebook |
| <b>LPG (propane–butane)</b> | 63.1  | 0.227 | National rulebook |
| <b>Electricity</b>          | 254.1 | 0.915 | National rulebook |
| <b>District heating</b>     | —     | 0.259 | National rulebook |

\* According to the Rulebook on Energy Audits, the CO<sub>2</sub> emissions is 0 kg CO<sub>2</sub>/kWh; however, according to the “IPCC Guidelines, 2006, National Greenhouse Gas Inventory, Volume 2: Energy,” a value of 0.403 kg CO<sub>2</sub>/kWh for wood/wood waste is indicated, which has been taken into account in the calculations.

### **Data protocols and limitations**

The methodology is primarily based on project-level data prepared by consultants in accordance with applicable national rulebooks and standard calculation procedures. Where multiple data sources are available or where information is incomplete, a consistent hierarchy is applied, giving priority to project-specific calculated values, followed by reported consultant documentation, and, where necessary, standard default parameters defined within this methodology. Default values are used only in cases where higher-quality or project-specific data are not available, in order to ensure consistency and completeness of programme-level reporting.

The methodology is subject to the following limitations:

- It relies on standardized assumptions rather than measured consumption data.
- Behavioural changes, rebound effects, and changes in user practices are not explicitly modelled.
- Reported energy savings represent engineering-based estimates relative to a standardized service level.
- Actual household energy bills may not decrease in all cases, particularly where comfort levels increase significantly after the intervention.

Despite these limitations, the methodology provides a transparent and robust basis for programme-level reporting and evaluation.

## 6. Impact Assessment Plan (Independent Verification)

### 6.1 Objective and Scope

To validate the reported results and ensure the technical quality of installations, an independent impact assessment will be conducted on a representative sample of the portfolio. This assessment aims to verify the physical existence of the assets, the quality of installation, and the actual operational parameters used by beneficiaries.

### 6.2 Sampling Strategy

#### 6.2.1 Outcome 1 (SME/Industrial)

"Production Normalization" Validating industrial energy savings requires correcting for production volume. A factory might show 50 per cent energy savings simply because it operated at half capacity.

- **Sample size:** Top ten largest loans by volume + five random SMEs.
- **Verification protocol:**
  1. **Asset verification:** Confirm equipment exists and is operational.
  2. **Production correlation:** The independent assessor must collect production data (e.g., "tons of product" or "operating hours") for the baseline year and the reporting year.
  3. **Specific energy consumption (SEC):** The validation will calculate the SEC (kWh/ton) pre- and post-investment.
    - *Pass Condition:* SEC must decrease.
    - *Fail Condition:* Total energy dropped, but SEC remained constant (indicating savings are due to production cuts, not efficiency).

#### 6.2.2 Outcome 2 (Households)

The assessment of Output 2 will target 10 per cent of the total beneficiary households. To ensure the sample is representative of the programme's impact and risk profile, a Stratified Random Sampling method will be used, based on the technological distribution observed in the portfolio:



- **Stratum 1: High impact (Heat pumps) – 65 per cent of sample**
  - *Rationale:* Heat pumps represent the largest share of energy (64 per cent and CO<sub>2</sub> savings. Verifying their seasonal efficiency (SPF) and usage patterns is critical for portfolio accuracy.
- **Stratum 2: High pollution risk (Biomass) – 20 per cent of sample**
  - *Rationale:* Biomass boilers carry the highest risk of particulate matter (PM) emissions if fuel quality is poor or operation is improper. Verification will focus on fuel moisture content and combustion settings.
- **Stratum 3: Passive solutions (Insulation/Windows) – 15 per cent of sample**
  - *Rationale:* These solutions provide long-term, unalterable savings but must be checked for installation quality (e.g., thermal bridging, gaps in insulation).

### 6.3 Data Collection Tools & Protocols

The independent assessor will utilize the following standardized tools during site visits:

- **Verification form - survey:**
  - **Asset Verification:** Photo evidence of the installed equipment (nameplate, serial number) to confirm it matches the loan application.
  - **Operational Check:** Confirm operating mode and setpoints; measure indoor temperature vs setpoint to assess potential rebound effects. (Scenario C).
  - **User Survey:** A short interview on comfort levels, change in monthly energy bills, and any technical issues encountered.
- **Technical Measurements (Spot Checks):**
  - **For Heat Pumps:** Measurement of outlet air/water temperature to verify operation mode.
  - **For Biomass:** Visual inspection of fuel storage (wood dryness) and flue gas color to estimate combustion efficiency.

### 6.4 Analysis Method

Data from the sample will be analyzed using a before-and-after comparison of verified energy consumption, where historical billing data are available, or a standardized engineering-based estimation approach, where such data are not available. The results will be used to derive a

programme-level realization rate, which will be applied retrospectively to adjust the final reported savings in the annual report.

## Conclusions and Recommendations

Based on the comprehensive review of the chronological registries, project documentation, and reported indicators for both Outcome 1 (SMEs) and Outcome 2 (Households), this validation concludes the following:

- **Data Integrity:** The reported results for energy savings and renewable energy capacity are considered technically valid and accurate. The internal consistency checks (Tier 1 and Tier 2) confirm that the aggregated figures fairly represent the physical reality of the implemented projects.
- **Methodological Robustness:** The "Implied Factor" analysis for industrial projects and the operational plausibility checks for households confirm that the reported impacts are within reasonable engineering limits.
- **Strategic Alignment:** The portfolio demonstrates a strong alignment with national decarbonization goals, with the verified results providing a solid baseline for the updated 2024–2027 targets.

### Outcome 1

#### Standardization of Portfolio-Level Data

The portfolio-level validation confirmed that reliable aggregation across heterogeneous industrial projects depends on consistent and well-structured data inputs. While the existing reporting framework performs adequately, further strengthening of standardization solutions would support long-term robustness as the programme continues to scale.

- **Validation Rule:** The reporting tool should maintain automatic range checks for key indicators (energy savings, CO<sub>2</sub> reductions, installed RES capacity), flagging values that fall outside expected portfolio-level ranges for targeted review.

#### Consolidation of the Portfolio-Level Methodology

The assessment and reporting of energy, climate, and environmental performance under Outcome 1 are based on a fully harmonized, portfolio-level methodology applied consistently across all supported industrial projects. The validation process confirmed that this methodology produces coherent and proportional results across energy savings, CO<sub>2</sub> reductions, and renewable energy capacity, despite the diversity of technologies and project scales.

Continued application of the existing framework, supported by periodic internal consistency checks at portfolio level, is considered sufficient to ensure credible reporting, target-setting, and scenario analysis throughout the remainder of the programme period.

## **Systematic Tracking of Co-Benefits**

Although energy savings, CO<sub>2</sub> reductions, and RES capacity remain the primary performance indicators under Outcome 1, the analysis demonstrates that industrial investments can also deliver measurable air-quality benefits, particularly in cases involving fuel switching or reduced on-site combustion.

- **Recommendation:** Where data availability allows, PM<sub>2.5</sub> and PM<sub>10</sub> reductions should be reported as standardized portfolio-level co-benefits, using conservative default factors to ensure comparability without increasing reporting burden for beneficiaries.

## **Steering the Technology Mix Toward Integrated Solutions**

The strong representation of renewable energy investments reflects high demand for on-site generation in the industrial sector, while the more limited share of stand-alone energy efficiency projects indicates remaining potential for demand-side optimization.

- **Recommendation:** Future programme phases should continue to promote integrated EE–RES investment approaches, supported by targeted technical guidance at early project stages. Strengthening the role of energy efficiency alongside renewable energy deployment can improve system sizing, enhance cost-effectiveness, and increase the resilience of industrial energy systems.

## **Outcome 2**

### **Standardization of Data Entry**

The validation process highlighted the importance of improved standardization within the reporting database to ensure consistency, comparability, and data integrity across reporting cycles. Variability in data entry practices, particularly for residential heating savings, can introduce outliers that are unlikely to be technically or operationally plausible and may distort portfolio-level results if not identified early.

To address this, future reporting cycles can incorporate automated validation rules within the reporting tool. These thresholds are not intended to reject entries automatically, but to prompt additional review and justification where reported values fall outside expected ranges. This approach supports early detection of potential data-entry errors while preserving flexibility for legitimate edge cases, thereby strengthening overall data quality without increasing reporting burden.

## Implementation of methodology

The assessment and reporting of energy and climate performance under the programme are conducted using a step by step and proportionate methodological approach, reflecting both data availability and the evolving scale of implementation. During the initial implementation phase, CO<sub>2</sub> reductions are calculated primarily at the level of individual technologies and investments, applying conservative, solution-specific emission factors. This approach ensures prudence and avoids overestimation at household level, particularly in the context of diverse heating technologies and consumption profiles.

As the programme portfolio expanded and aggregated impacts became increasingly relevant for strategic monitoring and reporting, the need for enhanced methodological consistency at programme level was addressed. In this context, a standardized weighted emission factor is applied within this report exclusively for programme-level analysis, target alignment, and indicative projections. The factor is used to align projected CO<sub>2</sub> reductions with validated energy savings, enabling a coherent interpretation of programme-level performance across indicators, while preserving the conservative treatment applied at the level of individual interventions.

Based on this structured approach, the report implements a programme-level CO<sub>2</sub> accounting framework that provides clear guidance for the consistent estimation and aggregation of emission reductions across technologies. The methodology defines standardized assumptions, reporting boundaries, and aggregation rules, and is aligned with national energy and climate reporting frameworks. The phased application of solution-level and programme-level methods reflects good practice in results-based programme management, ensuring that reporting frameworks evolve in line with programme maturity, data robustness, and policy relevance.

## Enhanced Monitoring of Co-Benefits (Air Quality)

Air pollution reduction is a major driver for the programme but has not been systematically tracked in the initial targets.

- **Recommendation:** Integrate the PM<sub>2.5</sub> and PM<sub>10</sub> calculation methodology (Section 2.4) into the standard reporting template. Every replacement of a solid fuel stove should automatically generate a "PM Savings" value based on the EMEP/EEA guidebook factors.

## Market Development for Under-Utilized Technologies

While heat pumps are performing well, envelope solutions (insulation/windows) and solar technologies are lagging, representing less than 20 per cent of installations.

- **Recommendation:** The Green Finance Facility to Improve Air Quality and Combat Climate Change in North Macedonia programme should consider a "Bonus Grant" or targeted marketing campaign for Envelope + Heat Pump packages. Improving insulation reduces the heating load, allowing for smaller, lower-capacity heat pumps to be installed, optimizing the overall investment cost and grid impact.

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Validation of energy, climate, and air quality results  
under the Green Finance Facility to improve air quality  
and combat climate change in North Macedonia

The analysis in the study “Validation of energy, climate, and air quality results under the Green Finance Facility to improve air quality and combat climate change in North Macedonia” verified the estimated energy savings, GHG emissions reductions, installed RE capacity, and air quality (PM2.5 and PM10) benefits for 2023-2025, covering SMEs and households. The programme achieved annual energy savings of 65,345 MWh in 2023-2025, approximately equivalent to the power consumed by 17,000 households annually. Emissions reductions of 50,701 tonnes of CO<sub>2</sub> per year have been achieved in the same period, equivalent to 11,500 passenger cars removed from the road.

The validation of the household investments confirms a strong preference for high-efficiency heat pumps, reflecting the clear market shift towards electrification. This technology is the primary driver behind energy savings, CO<sub>2</sub> reductions, and improvements in local air quality. Air quality co-benefits are significant: although not part of the original target framework, reductions in PM2.5 and PM10 emerge as supporting broader public health and environmental objectives.

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